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The Impact of Fiscal Policy on Inflation and Interest Rates in the UK using a Structural Vector Autoregression Analysis

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ABSTRACT

This study explores the interactions between government expenditure, real GDP, inflation, and interest rates in the UK using a SVAR model from 2004 to 2024. It was observed that government expenditure exhibits a reversal trend after a surge in spending and provides limited evidence on its impact on GDP. Inflation showed strong autoregressive movement, while interest rates were highly dependent on the previous values and respond to past fiscal actions and GDP shocks. The SVAR model provides evidence to support the idea that government expenditure shocks increase inflation and interest rates, while GDP's impact reduced inflation, suggesting productivity led growth. During economic crisis, GDP is the exogenous driver with expenditure and inflation increasing. Interest rates also rise in response to inflationary pressure. Residual diagnostics support a counter-cyclical fiscal accommodative monetary policy. The model indicates that fiscal policy can influence inflation and interest rates but its short-term impact on growth is weak

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1. Introduction

This paper looks at the short-term effects of the impact of government expenditure on output, inflation, and interest rates in the UK using a structural vector autoregression (SVAR) model. The study is built upon the aggregate demand-aggregate supply (AD-AS) framework, which provides a useful perspective to view the interplay between monetary and fiscal policy in reaction to macroeconomic shocks. In the short term, AD is the sum of finished goods and services produced within an economy. It consists of consumption (C), investment (I), government spending (G), and net exports (X-M), (AD = C+I+G+(N-X)). In the AD-AS framework, AD is a downward sloping curve showing the negative relationship between output and the inflation rate when all other variables are held constant. A fiscal expansion, like an increase in G, shifts the AD curve rightward. This leads to an increase in output (Y) and inflation in the short-term, shown by a rightward movement along the short run AS (SRAS) curve displayed in figure 1.

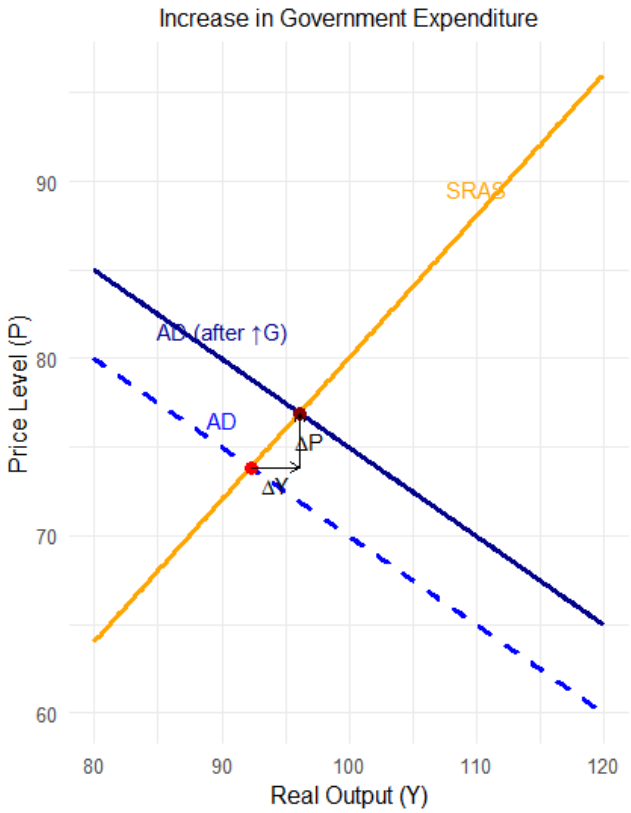

Figure 1

A central problem within the AD-AS frame is identifying the true impact of a government expenditure increase. As displayed in figure 1, a rise in AD increases both output and inflation in the short run. However, this effect is not distinct, as monetary policy responds endogenously

to changes in inflation and output, particularly in the essence of this study which looks at the UK who are an inflation targeting country. The Bank of England's (BoE) Monetary Policy Committee (MPC) are tasked by the government to retain inflation at the 2% level in the medium term. According to the Taylor Rule, a rise in inflation or output above potential, forces the central bank to raise interest rates which contracts aggregate demand (figure 2). The policy response from the bank creates a feedback loop that can alter the sole effect of fiscal policy on the macroeconomy. This creates an empirical identification problem: to quantify the impact of a government expenditure increase, it is necessary to separate fiscal policy from immediate responses from monetary policy reactions. This study uses a SVAR model to do so.

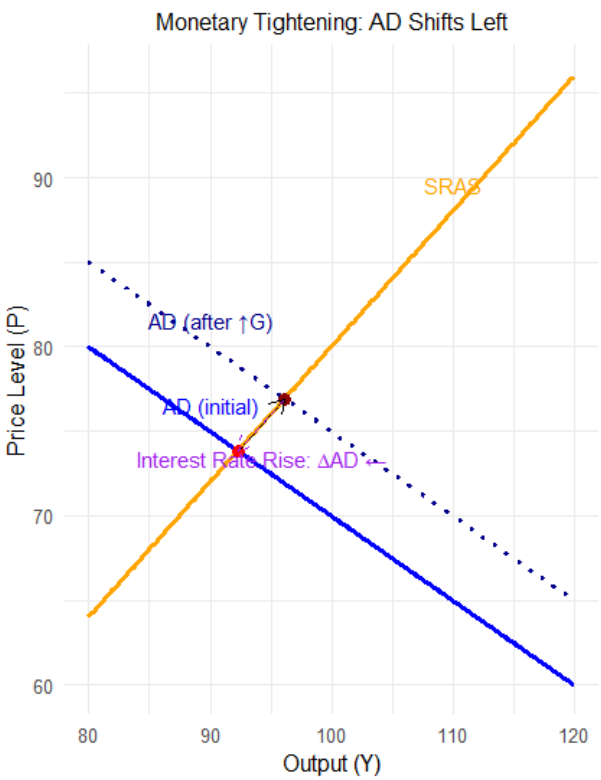

Figure 2

Although the focus in this study is on short-run dynamics, it is worth noting, an increase in government spending can also raise productive capacity. In this event, the long-run aggregate supply (LRAS) shifts to the right, evidenced in figure 3. This supply-side effect can moderate inflationary pressure and lead to higher sustainable output, but it is not the central concern of this paper which focuses on quarterly effects.

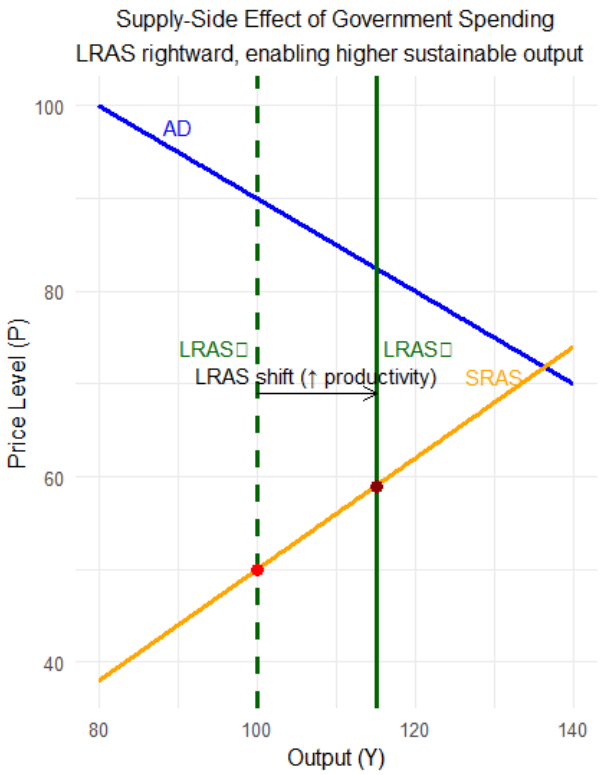

Figure 3

The identification issue is also at the centre of a greater theoretical argument between Keynesian economics and the Ricardian equivalence hypothesis. Keynesian's theory believes that fiscal policy is an effective tool for managing aggregate demand. Keynes (1936) observed in recessionary periods, private sector demand was too week, and without government intervention, the economy would remain in a recession. He argued governments should actively manage demand by increasing public spending, and/or cutting taxes during economic downturn to stimulate demand, or reduce spending, and/or increase taxes during periods when the economy is overheating to prevent inflation. Pontusson and Raess (2012) revealed this approach was employed following the 2008-09 Financial Crisis, where the UK economy experienced economic downturn. In December 2009, Prime Minister Gordon Brown introduced a temporary value added tax (VAT) cut, increased public infrastructure investment, nationalised parts of major banks, and enhanced the welfare state system. This multiplied into producing a stabilised economy and positive GDP growth by late-2010 (UK Parliament, 2010). However, the actions of Brown occurred during a low-interest rate period evidenced in appendix 1, which raised the question whether economic recovery was based on fiscal policy or monetary accommodation. Without separating the impact of government expenditure from interest rate changes, it is hard to establish the true impact of fiscal policy.

In contrast, is the Ricardian equivalence which was introduced by David Ricardo in the early 19th century and later formalised by Robert Barro in the 1970s. They proposed that government spending and taxation does not impact the overall economy. Barro (1974) argued when a government debt finances its spending, instead of using tax revenue, individuals expect future taxes will be needed to repay the debt and thus increase their savings. His study found that fiscal policy has no real effect on output, and governments attempt to stimulate demand is offset by private sector behaviour. While some studies support this view (Runkle,1991), others found that people do not aggregate their spending over time as consumption is more sensitive to income than consumption-smoothing model predicts (Cambell and Mankiw, 1989). This is due to liquidity constraints. Many low-income households do not have access to savings hence when they receive extra money, they spend it quickly because they must satisfy immediate needs. This is a theoretical debate of which this study does not attempt to solve, instead it contributes to it by using data to estimate the effect of a government spending shock on output, inflation and interest rates while accounting for the fiscal and monetary interactions.

The predictions from the AD-AS framework are tested empirically using an unrestricted VAR and recursively identified SVAR models. Based on New Keynesian theory, it is expected that a positive government expenditure shock will increase real GDP (output), inflation (price level), and interest rates (90-day yield). Results from the VAR provide a more nuanced view whilst the SVAR model provides support for these dynamics, especially the immediate inflationary and interest rate response. In contrast to real GDP which does not significantly react to lagged fiscal or monetary factors, expenditure seemed to show constant growth and shows limited impact on output. Inflation was primarily driven by its own past values, and while government expenditure and GDP both had positive coefficients, their impact was statistically insignificant. Interest rates were highly autoregressive and seem to react more to historical GDP and inflation than to shocks in expenditure. The SVAR's impulse response functions (IRFs) show that a government expenditure shock leads to a statistically significant rise in inflation and interest rates in the short run, while the effect on output was positive but statistically less significant. These findings highlight the limited impact of fiscal policy on real activity, and the dominant role of inflation and BoE reaction in shaping the short-run UK macroeconomic outcome.

This paper contributes to the broader literature on the use of the SVAR model in fiscal policy analysis between 2004-2024, a series which included various economic shocks impacting the UK economy in several different ways. The results showed that while government expenditure

shocks have distinct and statistically significant short-term effects on interest rates and inflation, their impact on real output is less strong. In the context of the UK, fiscal stimulus creates higher prices. Although this matched the theory's prediction, an explanation for the lack of effect on real output is down to the UK being an inflation-targeting, capacity-constrained economy where monetary policy and expectations adjust quickly. This causes prices and interest rates to respond more quickly and aggressively than real activity leading to a transitory and muted rise in output. The finding adds to the ongoing debate around the effectiveness of fiscal policy under modern frameworks, suggesting that without addressing supply-side constraints, fiscal policy alone is unlikely to deliver sustained growth.

2. Literature Review

The empirical literature examines how government expenditure effects inflation, interest rates, and output. Across the studies there appears to be mixed findings, but this is largely due to differences in country, time periods, modelling approach, and economic structures. Although some research supports Keynesians predictions, others challenge it, noting the role of expectations, monetary policy responses, and structural constrains. This literature review uses a range of studies to provide a solid foundation for interpreting the effects of fiscal policy on the macroeconomy in the UK.

2.1 The Effect of Government Spending on Inflation

Khundrakpam and Pattanaik (2012) analysed the relationship between government expenditure and inflation in India over both 1953-2009 and 2008-2010, during the financial crisis response. A rise in expenditure during the 2008-10 period did not significantly contribute to inflation, consistent with the Ricardian hypothesis equivalence. This was due to weak private sector demand. For the fiscal stimulus to work, it requires additional spending in other parts of the economy to boost growth. This meant that the fiscal stimulus offset the missing consumption and investment caused by the crisis, instead of creating excess demand. However, in the long run between 1953-2009, the study found a statistical long-run relationship between the two. A 1% increase in spending was estimated to increase inflation by 0.25%. This is because when the government financed the deficit by borrowing from the central bank it created an increase

in reserve money. As the Indian government runs persistent deficits and rely on monetary financing, it increases the money supply which can impact the level of inflation.

Wang (2023) examined the long-run relationship between government defence spending and inflation in the UK using a wavelet analysis. The study used historical data from 1715 to 2019 and found that an increase in expenditure contributed to inflation at 8-to-48-year cycles between 1890 and 1940, and at 50 to 65-year cycles between 1790 and 1860. The rise in inflation, drove increases in defence expenditure in 7-to-20-year cycles between 1840 and 1870, and again during the 1940s and 1980s. This bidirectional relationship shows interesting insight into the time-varying and cyclical nature of fiscal-inflation dynamics. Hartley's (2016) previous results support these findings highlighting how defence inflation outpaces general price indices, showing upward pressure on budgets however, Wang notes the need for more robust simulation-based significance testing to confirm them. Hartley's study used a traditional cost analysis and index-based comparison and reports findings in a descriptive manner based on cost escalation trends in operations. Nevertheless, Wang study reveals the inflationary effects of government spending do not act in accordance constantly through time, challenging the simple linear models. The findings suggest that the inflationary consequence caused by the increase in government spending may be dependent on the current state of the economy and policy regimes.

2.2 The Effect of Government Spending on Interest Rates

Yong and Dingming (2019) explored the impact of a news shock about US government expenditure on interest rates and investigated the role of agent's expectations, and their contribution to the variance of the outcome. Their study employed a large Bayesian vector autoregressive (BVAR) model using quarterly data from 1974 to 2018 and found that the effects of an expenditure surprise shocks on interest rates were unclear. News about increases in expenditure caused significant rise in both short and long-term interest rates. More importantly, "news" shocks explain up to 10% of the variation in the 10-year Treasury yield, a significantly stronger effect than the unexpected expenditure shocks. Once the study used structural models to detangle changes in long-term and risk-neutral interest rates, and term premiums, they concluded that a rise in interest rates was primarily due to increased expectation of future monetary tightening. These findings imply that the market does not wait for the government to

spend money, they react to announcement. As a result, the market expects the bank to raise interest rates to prevent inflation due to expected rise in expenditure, causing yields on government bonds to rise immediately.

Fadol (2024) analysed the effect of a government spending shock on macroeconomic variables in the Sudanese economy. The study took annual data from 1989 to 2019 and used a Dynamic Stochastic General Equilibrium (DSGE) model and SVAR approach to ensure that the theory matched with the data and the quality of the econometric model description varies. The results showed that government spending had a negative impact on the selected macroeconomic variables. Specifically, inflation and the exchange rate did not respond positively to government spending shocks which contradicts Keynesian theory that predicts an increase in aggregate demand through government spending should raise prices and potentially depreciate the exchange rate. A possible explanation for this is that the Sudanese monetary policy tools were not strong enough to manage the exchange rate. Therefore, when the government spends more, the effects do not multiply. The authors recommend that fiscal and monetary policy should be coordinated so that they both work together to either stimulate the economy or cool the economy as it would better suit the Sudanese economic structure.

2.3 The Effect of Government Spending on Output

Glocker et al. (2019) assessed the impact of a UK government consumption shock on output, private consumption and investment using a time-varying parameter factor augmented vector autoregressive model (TVP-FAVAR) over the period 1966:1 to 2015:4. They aimed to investigate how government spending multipliers have evolved over time, with a focus on their cyclical behaviour. The study found that fiscal multipliers exhibited a cyclical variation over time. Specifically, multipliers are above one during recessions and below one during economic booms. This indicates that government spending is more effective at stimulating output when the economy is weak and has little impact during periods of economic growth. This cyclical variation is consistent with financial frictions and economic slowdowns as key drivers of effective fiscal policy. The work of Lee et al. (2020) also supported this pattern, finding that in US data, cumulative multipliers are significantly higher and exceed unity during high unemployment. Beyond cyclical influences, Glocker et al, also examine structural factors. However, they find no evidence that multipliers are higher during the 2009-2015 zero lower

bound (ZLB), contrary to some of the findings in the literature (Klein and Winkler, 2021). This divergence may reflect differences in monetary policy expectations, duration of the ZLB episode, or the presence of unconventional monetary policy tools that partially offset fiscal stimulus. It suggests that the ZLB is not enough on its own to guarantee fiscal effectiveness. Though the analysis highlights the importance of cyclical drivers, it finds that structural policy changes play little role in explaining the variation in multipliers.

Ciaffi et al. (2024) analysed increases in government spending effect on GDP and the public debt-to-GDP ratio and whether the effects differed depending on the level of existing public debt. The authors took a sample from 14 Organisation for Economic Co-Operation and Development (OECD) countries from 1981 to 2017 using a Local Projections (LP) and Smooth Transmission LP model to estimate impulse response functions (IRFs). Following the methodology of Blanchard and Perotti (2002), they use a structural VAR approach to identify fiscal policy shocks assuming that government spending is exogenous within a quarter. The work of Caffie et al found that increasing government spending raised output persistently, confirming Keynesian effects, and that fiscal multipliers are positive and often greater than one. In linear models, higher government spending led to a reduction in the public debt-to-GDP ratio, aligning with a previous study (Auerbach and Gorodnichenko, 2012), which contradicts the assumption that spending increases debt sustainability. In non-linear models, fiscal debt tends to rise more when initial debt levels are already high, which highlights the significant role of fiscal policy in high-debt scenarios. This contrasts with Auerbach and Gorodnichenko (2012) where lower debt improved fiscal effectiveness. The authors explain that when the fiscal multiplier exceeds the inverse of the public debt-to-GDP ratio, stimulus can reduce the debt ratio even if debt is high. This is because as debt rises, the multiplier needed to reduce the debt ratio becomes smaller, making stimulus more effective more likely in high debt scenarios.

2.4 Summary of the Existing Literature

Table 1: A review of the empirical studies on government spending on inflation, interest rates and output

Author/ Year	Country(ies)	Period	Methodology	Conclusion

Khundrakpam and Pattanaik (2012)	India	1953-2009 and 2008-2010	ARDL	There is an impact between expenditure and inflation in the long run but no short-term effects.
Wang (2023)	UK	1715-2019	Wavelet analysis	An increase in government spending contributed to inflation.
Yong and Dingming (2019)	US	1966:1-2015:4	BVAR	““news” shocks explain up to 10% of the variation in the 10-year Treasury yield.
Fadol (2024)	Sudan	1989-2019	DSGE and SVAR	Expenditure had a negative impact on the selected macroeconomic variables
Glocker et al. (2019)	UK	1966:1-2015:4	TVP-FAVAR	Fiscal multipliers are above one during recessions and below one during economic booms
Ciaffi et al. (2024)	14 OECD	1981-2017	Smooth Transmission LP, IRF, and SVAR	Fiscal multipliers are positive and often greater than one

The empirical literature discussed above suggests that the relationships between government spending, and the macroeconomic variables produced mixed findings. The sorts of data, the period that was considered, the alternative econometric methods, and the various characteristics of the range of countries, contributed to the inconclusive results. However, by focusing on the studies based on the advanced economies, it provides a base understanding of what is to be expected throughout this research on the UK.

2.5 Limitations in the Existing Research

Fadol (2024) explored the impact of government spending on interest rates using a DSGE model and complimented this later with a SVAR approach on the Sudanese economy. However, DSGE model assume agents have perfect information and rational foresight about consumption, savings and investments over their lifetime. This assumption does not hold. Households face uncertainty about future income and policy changes that may affect inflation and interest rates. For example, in 2020 the UK government increased spending to support businesses and households due to the COVID-19 pandemic. Following this, it was evident that households did not react in a "rational" manner as many increased savings because of uncertainty, while others used the money to cover debt and basic expenses. At the same time, inflation rose more sharply than expected, and interest rate expectations shifted rapidly. These short-term market responses driven by uncertainty and liquidity constraints would not be captured accurately using either model. Although Fadol later complimented his study using a SVAR, it was based on the Sudanese economy. In a developing economy like Sudan, data is often incomplete or subject to large revisions due to political instability. This reduces the reliability of model estimates and increases the risk of producing misleading results.

Studies focusing on advanced economies face challenges. Glocker et al. (2019) applied a TVP-FAVAR model to the UK finding that the effect of government spending on output varied cyclically being larger in recessions than booms. Then, Wang (2023) used a Wavelet analysis in the UK to find a positive cyclical link between government defence spending and inflation. Finally, Ciaffi et al. (2024) utilised the LP and SVAR models to conclude that government spending raises output. Though each study uses different methodologies and data scopes, they all share a common limitation: the challenge of reliably identifying causal effects of government spending due to the reliance on structural assumptions and model sensitivity. The FAVAR, wavelet, and SVAR-LP models all depend on how fiscal shocks are specified and on the assumption that those shocks are exogenous to other macroeconomic dynamics, which is difficult to guarantee in the real-world setting. For instance, government spending might respond to output or inflation in ways not fully captured by the models. Additionally, the models often assume linear or smooth non-linear transmission mechanisms, but in practice, fiscal policy effect can be state dependent. In Wangs case, the use of historical data going back to the 18th century further complicates identification due to quality data concerns and structural breaks in the economy. While the results provide useful insights into the dynamics of fiscal

policy, they may not reflect the true causal impact of government spending under changing economic conditions.

3. Data

This section will outline how monetary and fiscal policy is implemented in the UK. Next, it will describe the data used throughout this study on government expenditure, real GDP, inflation, and interest rates. It will conclude with summary statistics for all the variables to allow for the assessment of the average level and volatility in the data.

3.1 Implementation of Monetary and Fiscal Policy in the UK

Fiscal and Monetary policy are handled by distinct organisations in the UK, each with the specific responsibility to maintain economic stability. Fiscal policy is managed by the Chancellor of Exchequer, operating within His Majesty's Treasury (HMT), who are responsible for implementing and announcing fiscal measures. In January, the Chancellor is required by the Charter for Budget Responsibility to address any fiscal changes they wish to implement for the Spring Budget, which is delivered in March. Once HMT, HM Revenue and Customs (HMRC), and the Department for Work and Pensions (DWP) agree on a timetable, the Office for Budget Responsibility (OBR) produce an economic and fiscal outlook based on the cost of the Chancellor's policy. In March, the budget is presented to the House of Commons outlining its plan, while the OBR simultaneously publishes an updated outlook to inform debates regarding the budget. As the budget speech is announced, the OBR publishes an updated forecast, allowing for the credibility of proposed fiscal plans. The budget is debated in Parliament over several days, and once approved, HMRC and DWP implement the changes which could be from March to June or beyond, depending on the complexity. The OBR monitors the fiscal change outcome against their projections and HMT reports to Parliament regarding any updates in public sector finances, particularly at the start of the next financial year in April, and during the Autumn statement in November when the next OBR forecast is usually published.

The implementation of government expenditure is a long process that cannot react immediately to real GDP, inflation, or interest rates in the short term due to the procedures it must follow. This study examines the contemporaneous impact of variables following a shock whereby a

variable can only effect those placed after it, but not to those placed before it, hence why expenditure is placed first in the list below. Real GDP can respond instantly to government expenditure, but expenditure cannot react to GDP simultaneously. Next is inflation which is an outcome from demand-side pressures therefore, it reacts to GDP and expenditure but cannot impact GDP in the same quarter. Finally, the interest rate. The MPC of the BoE meet 8 times per year, unlike other fiscal institutions that are restricted by fixed schedules. The MPC forecasts and uses timely economic data to guide their decision making. To assist the Bank's mandate to remain inflation at its target level, monetary policy can be implemented in days therefore, it is the only variable that can fully respond within the quarter to movements in other variables, explaining why it was place last in this order.

1. Expenditure
2. Real GDP
3. Inflation
4. Interest Rate

However, during crisis periods institutional timing constraints in fiscal policy can temporarily loosen. Ministers have the power and authority to implement directional fiscal measures within weeks to by-pass the standard budgetary calendar through emergency orders. The National Audit Office (2020) revealed the UK Treasury rolled out support schemes within the same quarter to the COVID-19 pandemic. This means that in a crisis setting, the need for government expenditure responds immediately from a shock in output, changing the order to follow:

1. Real GDP
2. Expenditure
3. Inflation
4. Interest Rate.

In a crisis, GDP falls sharply first which triggers an immediate need for government intervention. As in this case fiscal rules are relaxed, expenditure responds within the same quarter to stabilise the economy. Following this, demand begins to recover, causing potential inflationary pressures, leading the bank to adjust interest rates. This illustrates that the "no within-quarter response" assumption can be violated during crises.

3.2 Government Expenditure

Government expenditure is the total consumption expenditure (P3) Chain Volume Measured (CVM), Seasonally Adjusted, reported by the ONS who are the primary data source provider in the UK (ONS, 2025). This is the amount of money that public sector corporations spend on education, healthcare, defence and welfare. It excludes transfer payments like benefits and pensions, focusing only on actual consumption. The CVM represents adjustments for inflation so that the data captures real expenditure behaviour.

3.3 Real GDP

Real GDP seasonally adjusted is the total output in an economy which removes the effect of price change over time to show actual growth in production, while accounting for predictable seasonal changes that occur every year. This provides a better comparison of GDP from quarter to quarter, measured in chained quarterly terms in millions of pounds (ONS, 2025). Again, this information was sourced from the ONS which is an independent body. This means it is not subject to political interference regarding the statistics it produces on the economy.

3.4 Inflation

Inflation is the average level that price increases over time which reduces the purchasing power of money held today as it requires households to spend more to buy the same amount of goods. This study uses consumer price index including owners' occupiers housing cost (CPIH) annual rate of all goods and services tracked. CPIH provides a more accurate metric for the cost of living where in the UK, housing costs take up a significant proportion of individuals incomes, especially in the private rental sector (ONS, 2024). However, the BoE (2024) revealed that they set its inflation target based on CPI. This means that monetary policy implications based on CPIH in this study should be made with caution. The inflation data used was obtained from the ONS. Many policymakers and institutions rely upon the figures produced by the ONS to make informed decisions based on the direction the economy is heading. The use of ONS data ensures inflation figures are up-to-date and provide an actual reflection of price change.

3.5 Interest Rate

The interest rate is a stated percentage of the borrowed or saved amount where it reflects the cost of borrowing money, or the return on saving. The BoE base rate is the most often cited rate in the UK. It affects both the returns on savings and the rate that commercial banks charge on credit, loans, and mortgages. For this study, the 90-day yield rate was used as a measure of the interest rates, as the bank's base rate does not actually represent the true cost of borrowing or saving to households or firms. Bordo and Haubrich (2024) suggests the 90-day yield rate is a market determined rate that incorporates the future of monetary policy and liquidity constraints. It captures how financial markets price short-term lending and borrowing, making it a more realistic indication of monetary conditions over time. The ONS does not publish information on the 90-day yield rate, therefore this data was sourced from the Federal Reserve Economic Data (FRED, 2025), which is controlled by the Federal Bank of St. Louis in the United States. The 90-day yield rate used in this study, is the UK 3-month Treasury yield which is published by the BoE but part of FRED's international data sets.

3.6 Summary Statistics

Appendix 2 is a summary of the statistics to give a descriptive overview of the variables used in this analysis between 2004:1 to 2024:4. Government expenditure shows a mean level of £21109.26 million with moderate variability of £52240.07 million. This represents the general upward trend in public expenditure over time. Real GDP exhibits a mean value of £563789.21, which is double that of government expenditures while capturing periods of economic growth and contractions. Inflation revealed a mean value of 2.66% which is quite close to the BoE target for most of the period, although its minimum and maximum suggest large fluctuations. Finally, interest rates showed an average level of 2.18% which is close to that of inflation. During the pre-2008 tightening phase to the low-interest rate environment after the crisis and throughout the pandemic, the interest rate fluctuated significantly.

4. Methodology

The main results from this study are derived using Sims' (1980) framework in RStudio. Sims critiqued the use of large-scale macro econometric models and instead proposed using VARs to analyse the relationships among variables in a data-driven way. Building on this foundation, the SVAR model used in this study estimates the isolated impact of structural fiscal expenditure shocks on real GDP, inflation, and interest rates. Through imposing a set of economically motivated restrictions on the SVAR, it becomes possible to identify the sole impact of an expenditure shock, free from external economic fluctuations. This approach transforms the correlated residuals into IRFs, whereby the model traces the dynamic response of each variable to an expenditure shock, producing a visual graphic of the causal effects over time.

4.1 Variable Interactions

In the short run, an increase in expenditure causes a rise in real GDP. As noted previously, expenditure is a component of AD and according to Keynesian theory, an increase in any of these elements will raise GDP. Prices and wages are sticky in this case therefore, a rise in demand leads to an increase in production rather than higher prices. This causes a multiplier effect where more spending after an initial rise in expenditure increases GDP by a greater proportion. In the context of the UK this relationship was validated after the COVID-19 pandemic where the government increased spending on support programs which boosted real GDP growth. However, as output rises further, inflationary pressure emerges, especially when the economy is operating near full capacity. Excess demand arises when expenditure increases faster than the economy can produce goods and services. This results in demand-pull inflation whereby firms' respond to the situation by raising price instead of increasing supply. Therefore, higher government expenditure is an indirect cause to inflation. The rise in inflation triggers a response from the BoE. If inflation exceeds its target 2%, the Bank may cut back on buying government bonds or raise interest rate to reduce inflation and relax economic activity. By increasing interest rates, the Bank promotes saving and reduces borrowing, intended to reduce aggregate demand and inflation. This decrease in aggregate demand slows GDP growth helping to ensure price stability, which is referred to as the interest rate transmission mechanism of monetary policy. It shows the cyclical link between monetary policy and fiscal conditions. The

interactions between fiscal and monetary policy reveal a cyclical relationship. Interest rate decisions are also influenced by real GDP. When GDP exceeds potential level, wage growth accelerates, and the labour market tightens as firms face rising costs. This increases the risk of cost-push inflation. In anticipation, the BoE may raise interest rates to prevent the economy from overheating. Conversely, if GDP falls below potential, the Bank may lower rates to stimulate demand. But because this works with lags, the effect is not immediate. Interest rate raises may take household and firms some time to react, and it may take 1 to 2 years before the full impact becomes apparent. Therefore, the central bank must act before the data is announced and alter decisions accordingly.

4.2 SVAR Model Rationale

The interaction between expenditure, GDP, inflation, and interest rates interact with each other is interdependent, concurrent, and changes overtime. As seen previously, a rise in expenditure increases GDP and inflation in the short run, which causes a change in the interest rate. A change in the interest rate feeds through a change in expenditure and GDP. Therefore, it is evident that assessing the isolated impact of one variable on another becomes difficult to interpret as a change in one variable indirectly effects another. A simple regression method is unable to capture these changes as it assumes there is one directional effect on independent and dependent variables. However, expenditure, GDP, inflation, and interest rates have joint effects, influencing each other contemporaneously and with lags. The way these variables interact with each other violate the assumptions of the simple linear regression as using the model for the purpose of this study would result in inconsistent estimates due to simultaneity and endogeneity. Therefore, a SVAR was employed on this study to capture the effects.

A reduced form VAR model presents each variable as a function of its previous values and the previous of other variables. The purpose of this model is for forecasting, but because the errors are correlated through changes in variables having joint effects, it is not possible to assess the error components in a structured manner. For this reason, a SVAR was used. SVAR models apply economically motivated restrictions to distinguish structural shocks from the reduced form residuals (reduced form VAR). This eliminates the possibility of multiple simultaneous influences and allows the tracking of the causal impact of an expenditure shock over time on real GDP, inflation, and interest rates.

4.3 Identification Strategy and Variable Ordering

To find structural shocks in the SVAR model, a recursive identification technique strategy was applied using Cholesky's decomposition. This strategy forms a lower triangular structure on the contemporaneous interactions between variables. The ordering of variables placed on the SVAR assumes a variable can only affect those placed after in the system but not those placed before it. As stated previously, the ordering for this study is as follows:

1. Expenditure
2. Real GDP
3. Inflation
4. Interest Rate

Following Blanchard and Perotti (2002), government expenditure is the most exogenous variable whereby a shock to it can affect other variables in the same quarter, but a shock to those other variables does not impact expenditure in the same quarter due to implementation delays. Real GDP can respond in the same quarter to an expenditure shock which is consistent with Keynesian's multiplier, while inflation and interest rates shocks cannot cause GDP to react. GDP is an outcome from a change in policy and given the long process the policy must go through to be implemented, GDP changes cannot influence the policy in the short run, supporting the placement of GDP after expenditure. However, interest rates may react to expenditure shocks before changes in GDP are realised because of market expectations. An increase in government expenditure results in a larger fiscal deficit, which could indicate increased gilt issuance and higher inflation or short-term interest rates in the future. This can cause term premia, even without GDP data which would raise market interest rates upon announcement (IIzetzki et al., 2013). Nevertheless, inflation responds simultaneously to shocks in expenditure or GDP, but not to interest rates. Inflation is an outcome of demand-side pressures therefore, it reacts to GDP and expenditure. Inflation cannot impact GDP in the same quarter due to the recognition lag. Economic data is published with a delay, meaning a rise in inflation cannot impact present GDP in the same quarter due to the inflation data representing previous price levels. This lag justifies why inflation is placed after GDP and expenditure. Lastly, shocks to the other three variables will have an immediate effect on the interest rate. This ordering reflects the way monetary and fiscal policy is implemented in the UK and what

is commonly used in the literature (Mountford and Uhlig, 2009). It is important to note that the results from this are conditional on the ordering employed. Though it has been commonly used throughout previous studies, alternative ordering could be explored to validate the findings.

5. Empirical Results

5.1 Stationarity Testing

The test for stationarity is an important concept when assessing time series data. A series is stationary if the mean and variance do not change over time, and there is no correlation between previous values and the time which the series is identified. If any one of the three conditions fails to be met, it would indicate non-stationarity. Granger and Newbold (1974) warned spurious results may arise when conducting regression analysis on non-stationary a time series, especially when the series shares common trends or seasonal patterns. This can result in misleading t-statistics or a high R-squared value even when no significant relationship exists. To check for stationarity, the Augmented Dickey-Fuller (ADF) and Phillps Peron (PP) test was used on the series in Figure 1. These results are presented in Table 2.

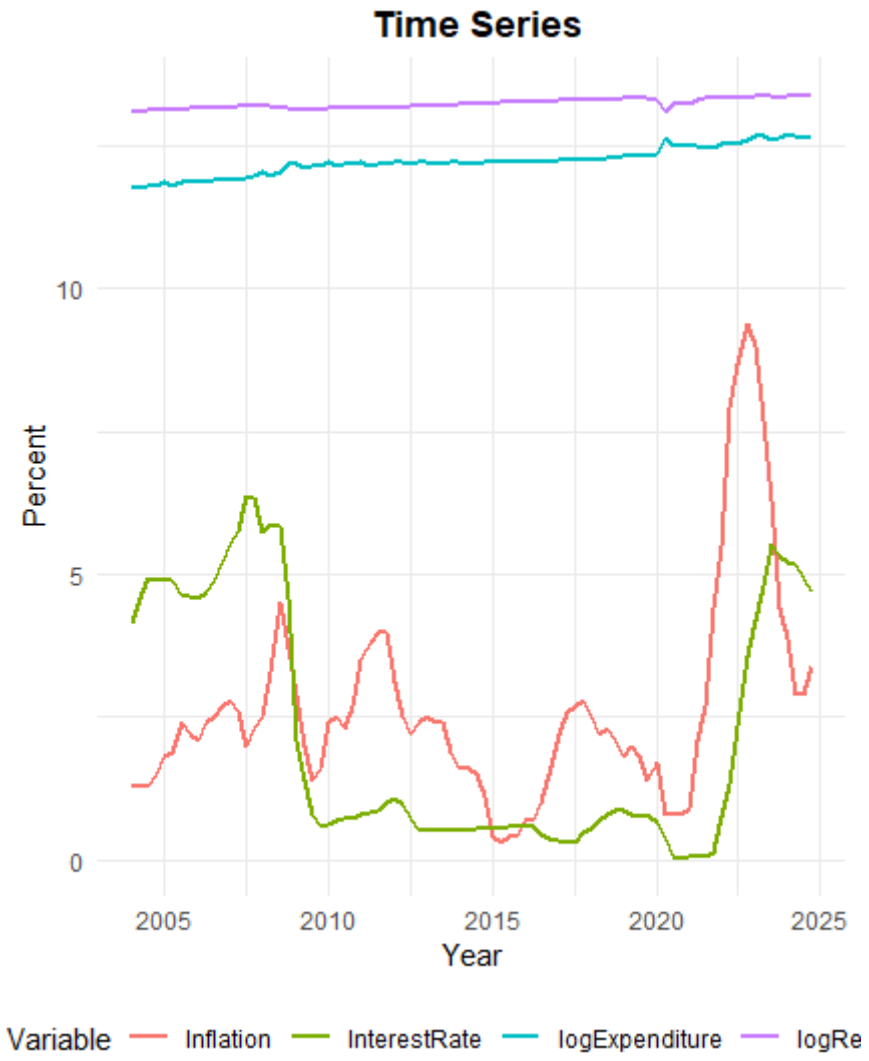

Figure 1: Time Series (2004:1 – 2024:4)

Table 2: Results of ADF and PP tests

Variable	ADF Test			PP Test		
	t- statistic	p-value	Conclusion	t- statistic	p-value	Conclusion
Log-Expenditure	-1.8551	0.6353	Non-stationary	-14.385	0.2701	Non-stationary
Log-Real GDP	-3.2936	0.07826	Non-stationary	-32.603	0.01	Stationary
Inflation	-2.4586	0.3878	Non-stationary	-12.735	0.3677	Non-stationary

Interest rate	-1.7518	0.6777	Non-stationary	-2.5155	0.9527	Non-stationary

As all variables p-values are above the 0.05 threshold, we fail to reject the null hypothesis and conclude the series is non-stationary. Both VAR and SVAR rely on the assumption that the data used in the models are stationary. Therefore, the series was differenced to produce Figure 2. The graph shows the variables plotted against each other after subtracting each variable value from its previous value. To ensure that this series was stationary, the ADF and PP test was applied. Tests results are outlined in Table 3.

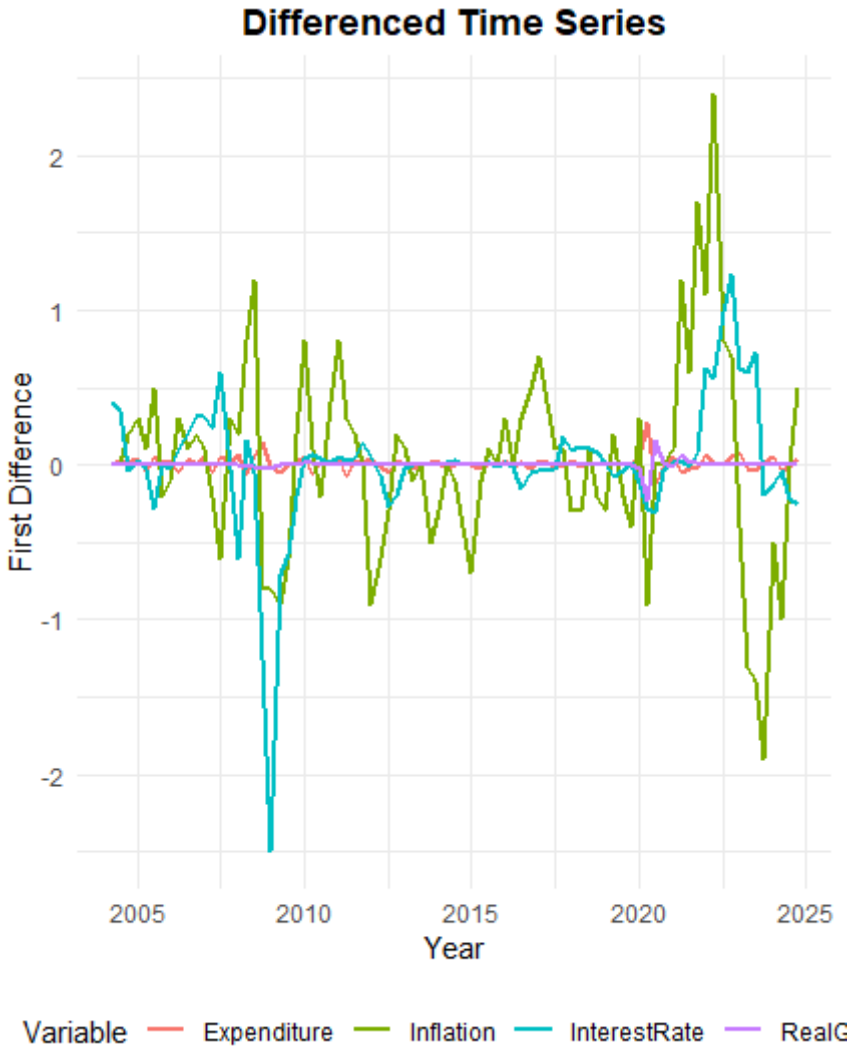

Figure 2: Differenced Time Series (2004:1 – 2024:4)

Table 3: Results of ADF and PP test on Differenced Series

Variable	ADF Test			PP Test		
	t- statistic	p-value	Conclusion	t-statistic	p-value	Conclusion
Log-Expenditure	-4.0252	0.01251	Stationary	-88.124	0.01	Stationary
Log-Real GDP	-4.8801	0.01	Stationary	-100.11	0.01	Stationary
Inflation	-4.1195	0.01	Stationary	-38.157	0.01	Stationary
Interest rate	-4.042	0.01174	Stationary	-30.307	0.01	Stationary

The outputs from the ADF and PP test on the differenced series confirms that all variables are stationary, ready for the model estimation. Where p value is 0.01 in Table 3, R studio returned a warning message saying that the p-value is less than printed p-value however, this is not of great importance as the variables still meet the stationarity criteria.

5.2 Lag Selection

Before estimating the VAR model, the test for the optimal lag selection was conducted. This is achieved by estimating VAR models using ordinary least squares for each equation across a range of lengths. The models were compared with various information criteria which included Akaike Information Criterion (AIC), Hannan-Quinn (HQ), Schwarz Criterion (SIC), and Final Prediction Error (FPE) in table 4. AIC penalises a model for having more estimators than it performs but assumes that reality has significantly more dimensions and that the model estimates are accurate. FPE is closely associated with AIC since both measures of fit and presume that none of the tests are true by testing the models on a different data set. However, SIC assumes that the correct model is one of the model orders. When it comes to penalising more parameters, the SIC and HQ are more stringent than the AIC. Ivanov and Kilian (2005) conclude that the AIC is preferred over HQ and SIC because it captures relationships through richer lag structures while cautiously avoiding overfitting. Therefore, the AIC is used in this study for the lag selection.

Table 4: Lag Order Selection Criteria Results

	AIC	HQ	SIC	FPE
Differenced Series	8	1	1	8

The AIC suggested a relatively high lag order. However, given the quarterly nature of the data, a lag length of 4 was selected for VAR and SVAR models, to remain consistent with the established literature (Blanchard and Quach, 1989; IMF, 2001). This ensures that there are enough lags to capture the different effects between the variables while preserving enough degrees of freedom.

5.3 VAR Results

Beginning with expenditure in table 5, the first two lags of expenditure are statistically significant at the 1% level, exhibiting negative coefficients. This reveals a surge in expenditure is reversed over the following quarters. These kinds of dynamics are common in government fiscal behaviour, where a spending surge is a response to a short-term shock. It acts as a stabiliser to push the economy towards its long-term fiscal goals. While the third lag was negative, it was not significant at the 10% level, and fourth the lag was slightly positive but remained insignificant. Looking at real GDP, the first three lags are significant at the 5% level and returned negative coefficients. This indicates high GDP in previous periods is associated with a lower current expenditure which could reflect a counter-cyclical fiscal policy imposed by the government to stop the economy from overheating. Focusing on inflation, only the first lag is statistically significant at the 5% level showing a small positive effect on current expenditure. This implies that rising prices may lead to an increase in nominal spending in the short run because of inflationary pressure driving up overall expenditure level. However, the remaining lags of inflation are mixed and statistically insignificant, showing no consistent delayed effects. Finally, interest rates. All lags were statistically insignificant with the coefficients being small with some positive and others negative. This suggests that interest rate movements in a four-period lag do not have a significant impact on government expenditure in this model. Overall, this regression explains that 39% of the variation in government expenditure, with considerable predictive power shown by the statistically significant F-statistic (p=0.005184).

For real GDP however in table 6, none of the lagged variables were statistically significant at any conventional level. The lagged coefficients of GDP magnitudes are small presenting both positive and negative signs, suggesting that there is not any significant autoregressive structure in the previous four quarters. Again, all lags of expenditure had positive coefficients but were statistically insignificant, which was surprising given the theory. Keynesians believe that government expenditure has a positive effect on output but, in this study, the evidence indicates expenditure is not significant enough to determine GDP. Both interest rates and inflation were statistically insignificant across all lags, meaning that there is no evidence to show that inflation or monetary policy has lagged effects on GDP in this model. Overall, the model's ability to explain real GDP is weak. With a r-squared (0.19), and adjusted r-squared (-0.019), the model preforms worse than a constant model considering the number of regressors. Furthermore, the F-statistic (0.9095) is statistically insignificant (p=0.562), confirming that the lagged variables does not explain real GDP in this model.

The model summary for inflation in table 7 showed past values of inflation are key drivers of current inflation. More importantly, the first lag is statistically significant at the 1% level and second at the 5% level, both with positive coefficients. This implies a strong autoregressive nature to the level of inflation such that rising inflation in previous quarters will persist into the current period, consistent with the trend observed in the raw data. Although the third lag is not significant its coefficient is negative, its coefficient is negative. The fourth lag is also negative but is statistically significant at the 5% level. This suggests another reversion after a period of inflation. Although, the lags for expenditure and GDP are statistically insignificant, all the coefficients remain positive. This reveals previous increases in expenditure or output could be inflationary but there is not enough evidence to support this statistically. The lagged coefficients of GDP are large, but they have equally large standard errors, indicating inaccurate estimates. All interest rate lags are statistically insignificant, meaning that in a four-period lag, interest rates do not have a direct impact on inflation. Overall, the model shows some explanatory power, with a r-squared (0.501) and adjusted r-squared (0.372), indicating that that the model explains 37% of the volatility in inflation. The F-statistic is highly significant ($p < 0.0001$) confirming that the lagged regressors together provide explanatory power. This explanatory power, captures a large proportion of inflation volatility, driven mostly by inflation's historical figures.

For interest rates in table 8, there is a strong autoregressive behaviour with the first lag being significant at the 0.1% level. This indicates that current interest is strongly influenced by its value the previous period, a pattern consistent with gradual adjustments made to bank rate by the MPC at the BoE. The second to fourth lags are small and not significant which indicates most of the autocorrelation is recorded at the first lag. Whereas GDP at the fourth lag had a huge negative and significant coefficient, while prior lags were also negative but not statistically significant. This suggests that lower interest rates are a result of poor economic activity during the past year, which reflects a delayed monetary policy reaction to sluggish growth. Expenditure at the first and fourth lag carry negative coefficients, with the latter being significant at the 5% level. These results indicate that higher expenditure from 4 periods ago is associated with lower current interest rates. This could indicate an accommodative policy in reaction to fiscal actions, or that fiscal expansions take place during economic downturn cause interest rates to decrease. All inflation lags were positive but not statistically significant. This means that there is no significant short-term pass through from inflation to interest rates. This could be due to delayed responses, or anchored inflation expectations. Overall, the model fits well with an r-squared (0.59), adjusted r-squared (0.48). The F-statistic ($p < 0.00000-1$) confirms that the regressors jointly contribute to explaining variations in the interest rate.

Table 5

	Estimation results for expenditure			
Variable	Coefficient	SD	t-statistics	p-value
Expenditure (1)	-0.680729***	0.195491***	-3.482***	0.000917***
Real GDP (1)	-0.541294*	0.264921*	-2.043*	0.045286*
Inflation (1)	0.024925*	0.011284*	2.209*	0.030890*
Interest Rate (1)	-0.011081	0.017879	-0.620	0.537678
Expenditure (2)	-0.628499**	0.218311**	-2.879**	0.005470**
Real GDP (2)	-0.647962*	0.303694*	-2.134*	0.036840*
Inflation (2)	-0.005952	0.012150	-0.490	0.625915
Interest Rate (2)	0.024425	0.021363	1.143	0.257295
Expenditure (3)	-0.366260	0.220367	-1.662	0.101552
Real GDP (3)	-0.641505*	0.308263*	-2.081*	0.041568*
Inflation (3)	0.002377	0.012658	0.188	0.851631
Interest Rate (3)	0.010545	0.021486	0.491	0.625318

Expenditure (4)	0.229179	0.191543	1.196	0.236062
Real GDP (4)	0.062737	0.280918	0.223	0.824013
Inflation (4)	0.016116	0.011463	1.406	0.164719
Interest Rate (4)	-0.006600	0.017970	-0.367	0.714651
const	0.031102**	0.010776**	2.886**	0.005356**

Table 6

	Estimation Results for Real GDP			
	Coefficient	Std. Error	t-Statistic	p-Value
Expenditure (1)	0.133	0.148	0.899	0.372
Real GDP (1)	-0.183	0.2006	-0.912	0.365
Inflation (1)	-0.0076	0.0085	-0.891	0.376
Interest Rate (1)	0.0175	0.0135	1.289	0.202
Expenditure (2)	0.201	0.1653	1.216	0.228
Real GDP (2)	0.0241	0.2299	0.105	0.917
Inflation (2)	0.0079	0.0092	0.857	0.395
Interest Rate (2)	-0.0104	0.0162	-0.644	0.522
Expenditure (3)	0.1144	0.1668	0.686	0.495
Real GDP (3)	0.1552	0.2334	0.665	0.509
Inflation (3)	-0.001	0.0096	-0.1	0.921
Interest Rate (3)	0.002	0.0163	0.125	0.901
Expenditure (4)	0.1069	0.145	0.737	0.464
Real GDP (4)	0.0442	0.2127	0.208	0.836
Inflation (4)	-0.0112	0.0087	-1.296	0.2
Interest Rate (4)	-0.006	0.0136	-0.443	0.659
const	-0.0027	0.0082	-0.327	0.744

Table 7

	Estimation Results for Inflation			
Variable	Coefficient	Std. Error	t-Statistic	p-Value
Expenditure (1)	0.9824	2.3199	0.423	0.6734
Real GDP (1)	2.6607	3.1439	0.846	0.4006

Inflation (1)	0.3769**	0.1339**	2.814**	0.0065**
Interest Rate (1)	0.1682	0.2122	0.793	0.431
Expenditure (2)	1.7034	2.5907	0.657	0.5133
Real GDP (2)	4.6429	3.604	1.288	0.2024
Inflation (2)	0.3072*	0.1442*	2.131*	0.0371*
Interest Rate (2)	-0.3821	0.2535	-1.507	0.1369
Expenditure (3)	1.4377	2.6151	0.55	0.5845
Real GDP (3)	5.1897	3.6582	1.419	0.161
Inflation (3)	-0.1143	0.1502	-0.761	0.4495
Interest Rate (3)	0.0537	0.255	0.211	0.8339
Expenditure (4)	1.6191	2.2731	0.712	0.479
Real GDP (4)	2.3168	3.3337	0.695	0.4897
Inflation (4)	-0.2724*	0.136*	-2.002*	0.0496*
Interest Rate (4)	-0.1752	0.2133	-0.822	0.4145
const	-0.0870	0.1279	-0.680	0.4989

Table 8

	Estimation results for Interest Rate			
Variable	Coefficient	Std. Error	t-Statistic	p-Value
Expenditure (1)	-2.6667	1.3995	-1.905	0.0614
Real GDP (1)	-2.0419	1.8966	-1.077	0.2858
Inflation (1)	0.1274	0.0808	1.576	0.12
Interest Rate (1)	0.694***	0.128***	5.422***	<0.001***
Expenditure (2)	0.1218	1.5629	0.078	0.9381
Real GDP (2)	0.7582	2.1742	0.349	0.7285
Inflation (2)	-0.1103	0.087	-1.268	0.2094
Interest Rate (2)	-0.0931	0.1529	-0.609	0.5449
Expenditure (3)	-0.8996	1.5776	-0.57	0.5706
Real GDP (3)	-0.4139	2.2069	-0.188	0.8518
Inflation (3)	0.0612	0.0906	0.676	0.5019
Interest Rate (3)	0.114	0.1538	0.741	0.4613
Expenditure (4)	-3.4671*	1.3713*	-2.528*	0.014*

Real GDP (4)	-4.4906*	2.0111*	-2.233*	0.0292*
Inflation (4)	0.1024	0.0821	1.248	0.2168
Interest Rate (4)	-0.0798	0.1287	-0.62	0.5373
Const	0.0873	0.0771	1.131	0.2623

5.4 Covariance and Correlation of Residuals

The diagonal elements of the residual covariance matrix in figure 3 are the variances of the residuals for each equation in the model. They indicate the proportion of the variability in each variable that remains unexplained by the model. Lower residual variance suggest that model captures dynamics of that variable well, while higher values indicate poor model fit.

	logExpenditure	**logRealGDP**	**Inflation**	**Interest Rate**
logExpenditure	0.00193	-0.00119	-0.00497	-0.0045
logRealGDP	-0.001188	0.001106	0.00517	0.002706
Inflation	-0.004965	0.005170	0.271759	0.04791
Interest Rate	-0.004498	0.002706	0.04791	0.098902

Figure 3: Residual Covariance Matrix

In the case of expenditure (0.00193) and GDP (0.001106), exhibit low variances suggesting that the model fits the data well. However, with inflation (0.271759) and interest rate (0.098902) residual variance is high in comparison to that of expenditure and GDP. This could indicate that there are other variables not included in the model that may account for their residual differences.

The off-diagonal elements in the matrix show how a shock in one variable is linearly associated with a shock in another. Since the covariance is symmetric, a shock to one variable has a linear effect on the other.

There is a negative covariance between expenditure and real GDP (-0.001188) which implies that an unexpected rise in expenditure results in a decline in GDP, and vice versa. This suggests a substitution pattern, showing that a shock in government expenditure crowds out private investment. Similarly, a negative covariance between expenditure and inflation (-0.004965) indicates that an unexpected rise in expenditure leads to a decrease in inflation, a finding contrary to the macroeconomic literature, which suggests expenditure shocks to be inflationary.

There is also a negative covariance in expenditure and interest rate (-0.004498). This implies that an unexpected shock in expenditure is associated with a decline in interest rates, which is observed in the data during economic crisis, when fiscal and monetary policy are expansionary.

However, there is a positive covariance between GDP and inflation (0.005170), which supports the previously mentioned notion of demand-pull inflation. There is also a positive covariance between GDP and the interest rate (0.002706), showcasing that due to an unexpected rise in GDP forces the BoE to tighten monetary policy. Finally, with the strongest off diagonal covariance is inflation and interest rate (0.047910). An unexpected rise inflation is strongly associated with an increase in the interest rate consistent with the BoE mandate to maintain price stability.

The residual correlation matrix in figure 4 standardises the values in the covariance matrix to allow for simple comparison of variable pairs regardless of their variances. All diagonal elements indicate a perfect correlation with each variable itself.

	logExpenditure	**logRealGDP**	**Inflation**	**InterestRate**
logExpenditure	1	-0.8129	-0.2168	-0.3256
logRealGDP	-0.8129	1	0.2982	0.2587
Inflation	-0.2168	0.2982	1	0.2922
InterestRate	-0.3256	0.2587	0.2922	1

Figure 4: Residual Correlation Matrix

The residuals of expenditure and real GDP (-0.8129) have a strong negative correlation. When expenditure deviates away from the model's expectation, GDP moves in the opposite direction. As aggregate demand is high in the economy, the government spends less, again supporting the idea of counter-cyclical fiscal policy. This reveals a short-term trade-off between output and consumption. There is a small negative correlation between expenditure and inflation (-0.2168). With an unexpected increase in expenditure, inflation rises slightly lower than expected. This may be because the impact of increased spending takes time to show in the data. Between expenditure and interest rates (-0.3256), there is a moderate negative correlation. This is consistent with the idea that higher government spending is associated with lower interest rates used to stimulate the economy during economic downturns. Again, GDP and inflation (0.2982) showed a positive correlation. There was also a moderate positive correlation between GDP and the interest rate (0.2587). This suggests that interest rate shocks are related in the

same and again may be a in relation to the how the BoE responds to the state of the UK economy. The residuals of inflation and interest rates (0.2992) have a positive correlation which is in line with monetary theory, where central bank increases rates to combat inflationary pressures.

5.5 SVAR Results

A recursive identification strategy was used to estimate the SVAR from the VAR, with optimisation done through scoring and direct methods. A log-likelihood value (-448.385) was obtained after 99 iterations of the model, which was calculated over a sample of 79 observations. The likelihood ratio (LR) test was used to jointly assess the over identification limitation on the structural A matrix, and the result shows a chi-squared statistic (1424) with four degrees of freedom. However, R studio suggested that the identifying constraints can be overly stringent or poorly defined in the SVAR, therefore care is adopted when interpreting the contemporaneous interactions.

The contemporaneous correlations between the four variables are determined by the estimated A matrix in figure 5, showing how each variable responds to structural shocks in other variables. The recursive structure assumes expenditure is the most exogenous variable and interest rate is highly endogenous, reacting contemporaneously to all other variables. The standard error of the coefficient in the A matrix is shown by each cell in figure 6. As part of Cholesky's decomposition strategy a value of 0 indicates that the relevant coefficient was not calculated

	logExpenditure	**logRealGDP**	**Inflation**	**InterestRate**
logExpenditure	1	0	0	0
logRealGDP	0.1897	1	0	0
Inflation	0.4647	-0.3048	1	0
InterestRate	0.3814	-0.0396	-0.1625	1

Figure 5: Estimated A Matrix

	logExpenditure	**logRealGDP**	**Inflation**	**InterestRate**
logExpenditure	0	0	0	0
logRealGDP	0.1125	0	0	0

Inflation	0.1145	0.1125	0	0
InterestRate	0.1259	0.1176	0.1125	0

Figure 6: Estimated Standard Errors for Matrix A

A rise in expenditure has an immediate positive effect on GDP (0.1897), with a t-statistic of 1.68 implying that fiscal multipliers are active in the same quarter but may be statistically weak. Inflation rises with expenditure (0.4647), as expected with a statistically significant effect (4.06). However, a rise GDP causes inflation to decrease (-0.3048), which was also seen to significant (-2.71). This negative effect is unusual as demand-side theory suggests higher GDP, results in greater demand, causing higher inflation. A possible explanation on why the finding differs from the theory is that if GDP growth is driven by increase productivity, greater output can occur without raising prices. Interest rates respond positively to an expenditure shock (0.3814) also being statistically significant (3.03), but do not respond as much to GDP shocks (-0.0396) with weak evidence to support this (-0.34). This may be because monetary policy focus is price stability whereby GDP, is a secondary concern. Interestingly, there is a negative coefficient between inflation and interest rates (-0.1625), which is not consistent with monetary policy behaviour but marginally significant (-1.44). This finding is counterintuitive and suggests there may be the need for restructuring in the model.

The estimated matrix B in figure 7 is the identity matrix, suggesting that every structural shock is assumed to have a contemporaneous impact on its related variable. This approach means that each shock is statistically independent from the impact of other variables. Therefore, the simultaneous interactions between variables are captured through matrix A in figure 5, and do no through cross-loadings in B, consistent with Cholesky's recursive identification scheme.

	logExpenditure	**logRealGDP**	**Inflation**	**InterestRate**
logExpenditure	1	0	0	0
logRealGDP	0	1	0	0
Inflation	0	0	1	0
InterestRate	0	0	0	1

Figure 7: Estimated Matrix B

The covariance matrix of reduce form residuals in figure 8 show the unexplained variability and contemporaneous co-movements of the model's residuals. The diagonal elements represent the volatility of residuals for each variable. It highlights that inflation shock is the most volatile (136.59), followed by interest rate (125.88), then GDP (103.6), and expenditure (100). This reveals that the nominal variables are subject to more uncertainty that real variables in this sample.

	logExpenditure	**logRealGDP**	**Inflation**	**InterestRate**
logExpenditure	100	-18.97	-52.25	-47.38
logRealGDP	-18.97	103.6	40.39	17.9
Inflation	-52.25	40.39	136.59	43.72
InterestRate	-47.38	17.9	43.72	125.88

Figure 8: Covariance Matrix of Residuals

The off-diagonal elements show how two variables move together simultaneously after a one standard deviation shock. An expenditure shock increases its own residual by 100 but has negative indirect effect on real GDP (-18.97), inflation (-52.25), and the interest rate (-47.38), which aligns with the earlier findings of counter-cyclical fiscal behaviour. GDP residuals show positive covariances on inflation (40.39) and interest rate (17.90), showing that growth shock raise inflation and triggers monetary policy to tighten. Inflation residuals are jointly move with interest rates (43.72) which is reflection of the central reaction with higher interest rates but shown through correlation. Finally, and interest rate residuals showed the second highest variance, and their positive co-movements with both GDP and inflation residuals reflects the BoE reaction function to the changing macroeconomy.

5.6 <u>Structural Impulse Response Functions</u>

To understand the effect of structural shocks over time, IRFs were generated by the SVAR. They chart the response over time to offer insight into both the contemporaneous effects and lagged dynamics. To show the effect of a single, one-time shock, SVAR was expressed as an infinite-order moving average process. The IRFs were plotted over a 10-quarter horizon to analyse the both the short-term and medium-term dynamics of fiscal shocks. Implementation also follows Lutkepohl (2007), including the computation of 95% bootstrapped confidence intervals to assess the statistical significance of the responses shown Appendix 6.

6. Discussion

After summarising VAR and SVAR results, several consistent patterns emerged across the models. Most notably, lagged values of government expenditure were statistically significant in explaining its own future values confirming a reversal trend after a period of significant distortion. This finding aligns with Barro (1979) work on the determination of the public debt. The government smooths expenditure over time to avoid large fluctuations in fiscal policy that could distort the economy. Large deviations away from the trend are to be corrected in future periods. This highlights fiscal authorities' long-term direction in stable expenditure, and reinforces assumption made in the SVAR model that expenditure is predetermined in a quarter and not driven by short term volatility unless during crisis periods.

For real GDP the analysis showed weak autoregressive behaviour, as all lagged variables were statistically insignificant with the model producing low explanatory power. This means that in the sample studied, expenditure, GDP, inflation, and interest rates offered little predictive power for explaining future changes in GDP. The fiscal policy impact of this suggests increases in government expenditure does not influence real GDP greatly limiting the effectiveness of delayed interventions. A possible explanation for this is that shocks in the sample period may have disrupted the transmission mechanism of fiscal policy. This reduces the effectiveness of fiscal policy when it is aimed at stimulating GDP even when government spending increases GDP, labour supply may be too constrained to respond proportionally. As a result, this breaks the causal relationship helping to explain the model's low predictive power. However, the SVAR model found an expenditure shock on GDP had a simultaneously positive effect on GDP, finding evidence for immediate fiscal multipliers. This shows that fiscal policy can be effective in the short-term, especially when it is implemented during the time of a recession. Similarly, Glocker et el. (2019) concluded that fiscal multipliers are more prevalent during sluggish economic growth, showing that the effectiveness of government expenditure is heavily dependent on the state of the economy. Therefore, the immediate impact of fiscal expansion suggests that well timed government spending remains a vital tool for stabilising GDP following a demand-side shock.

In the estimated VAR model, inflation was not statistically dependent on previous values of expenditure, but all the coefficients were positive. This suggests that an increase in government expenditure can increase inflation but with weak effect. Conversely, the autoregressive

movement of inflation showed positive coefficients on the first 2 lags. This indicates that previous inflation it better at predicting current inflation than government expenditure or the remain variables in the model. Despite GDP having a large positive coefficient on inflation, especially at lags 2 and 3, past changes in GDP did not have a clear predictive power for inflation in the UK during the period studied. The positive relationship between output and inflation is consistent with the Phillips Curve framework but the lack of statistical significance in the VAR analysis, highlights a potential limitation of such frame applied to the real world. The SVAR model showed that a rise in expenditure causes an immediate increase in inflation which was seen to statistically significant, highlighting positive short-term fiscal multipliers on prices. This finding contested that of Khundrakpam and Pattanaik (2012), who discovered that an increase in government expenditure had no immediate impact on inflation. Khundrakpam and Pattanaik's utilised the ARDL model which focuses on equilibrium relationships in the long run. The model they used does not account for isolated shocks meaning that an immediate inflationary response from an expenditure shock could be diluted by other movements in the data. However, this paper did find that an expenditure shock four lags ago had a positive impact on inflation, which aligns with the findings of Khundrakpam and Pattanaik long run relationship between expenditure and inflation. This suggests in the UK context, government expenditure can have both immediate and delayed inflationary effects, but the strength and timing of these effects depend on the economic environment and the methodological approach used to identify them. Therefore, fiscal policy influence on inflation is not uniform and must be understood through model's that can capture short-term and longer-term dynamics.

The impact of government expenditure on interest rates in the VAR model was weak in the short run. Only lag 4 was seen to be statistically significant with a negative coefficient. That means that an increase in government expenditure four quarters ago caused a decrease in the current interest rate, possibly reflecting a lagged accommodative monetary policy response to fiscal expansion. This pattern is consistent with the behaviour of monetary authorities during economic downturns, where high government expenditure is often accompanied by lower interest rates to stimulate aggregate demand. However, this inverse relationship is not present in the shorter lags, and most coefficients are not statistically significant, suggesting the effect is not immediate, but potentially delayed. By contrast, the SVAR analysis revealed that government expenditure had a statistically significant immediate positive effect on interest rates. This indicates that the BoE responds to unexpected increases in government spending by raising interest rates. It suggests that the shocks are not fully neutralised by monetary

authorities, especially when viewed through a structural lens. This conclusion aligns with Young and Dingming (2019) who found that the expectation of an increase in government spending before it occurs can raise interest rates. Their finding emphasises the role of expectation channels, where the anticipation of fiscal expansion drives expectations of tighter future monetary policy, leading to a rise in current yields. These findings support Keynesian view that the central bank responds to fiscal shocks based on inflation expectations, while the delayed the delayed response in the VAR shows the importance of timing and policy lags. Nevertheless, the contemporaneous response of interest rates due to expenditure shocks observed in the SVAR is consistent with this expectation-driven mechanism. Moreover, this finding challenges the notion that fiscal and monetary policy are always aligned. It indicates a potential policy conflict when fiscal expansion pressures the monetary authority to pre-emptively tighten, reflecting a reaction function to geared inflation control. However, as mentioned previously, this study uses CPIH as the measure of inflation but the BoE targets inflation using CPI. The use of CPIH may slightly affect the timing of the observed relationships, because CPIH includes owners-occupiers housing cost, which tend to adjust more slowly than other components of inflation. Therefore, monetary policy responses that are linked to CPI might appear to lead or lag the inflation dynamics captured by CPIH, potentially introducing small distortions in the interpretation of policy effectiveness.

The VAR and SVAR results suggest that expenditure has important but opposing implications for UK imports and exports. In the VAR, delayed interest rate responses and a weak line between expenditure and GDP suggests a muted immediate effect on trade. The negative residual correlation between expenditure and GDP supports counter-cyclical fiscal policy, indicating that expenditure rises during downturns when import demand may be weak. However, in the SVAR, expenditure shocks raise GDP and inflation simultaneously, causing monetary policy to tighten via increasing interests. This could lead to currency appreciation, making exports less competitive, worsening the trade balance. Additionally, the surprising negative link between GDP and inflation in the SVAR, suggests supply side growth, allowing exports to rise without inflationary pressure. The data reveals that fiscal policy may boost imports in the short run, but its import on exports depends on the monetary response and whether output growth is demand or supply driven.

During the crisis period, a new recursive ordering was implemented in the SVAR model, where GDP was treated as the most exogenous variable followed by expenditure, inflation, and interest rates. This reordering reflects the assumption that in economic slowdowns, output

shocks are external, and fiscal and monetary policy are reactive. The results show a small positive contemporaneous impact of GDP in expenditure, indicated that fiscal policy response to the state of the economy, as previously hypothesised. Interestingly, GDP appears to reduce inflation, which may reflect productivity driven growth rather than demand side pressures. A large positive effect of expenditure occurred upon inflation, suggesting that fiscal expansions are inflationary during economic crisis. Additionally, expenditure also strongly influenced interest rates, indicating a monetary policy response to inflationary pressures generated by fiscal actions. However, the inflation to interest rate link remained negative as in pre-crisis model which may reflect either delayed central bank reactions or willingness to tolerate inflation to support short term recovery. The crisis period SVAR concludes that the government and central bank are responsive rather than proactive throughout the sample, which is consistent with the policy framework during recent crises episodes (National Audit Office, 2020)

7. Conclusion

This paper examined the impact of fiscal policy on interest rates and inflation in the UK using a quarterly time series over the period 2004 to 2024, utilising a SVAR model. The empirical evidence in this study showed mixed results. In particular, the VAR model showed that government expenditure follows a cooling period after a significant rise in spending path and is significantly influenced by GDP, suggesting a counter-cyclical fiscal stance. This suggests that fiscal policy has been used as a stabilisation tool, with higher expenditure during downturns and is lower during periods of growth. However, contrary to Keynesians expectations, expenditure did not exhibit a statistically significant impact on real GDP, which raises a policy concern regarding the effectiveness of fiscal policy stimulating output in the short term. Inflation was shown to be strongly autoregressive, with past values significantly influencing current levels, while GDP and expenditure had positive but statistically insignificant effects. Interest rates also demonstrated persistent autoregressive behaviour, with evidence that they respond to GDP and expenditure changes, which is consistent with when the BoE makes gradual changes to the bank rate. The SVAR also provided deeper insights. Government expenditure had an immediate inflationary effect, contrary to the notion of sticky prices. This suggests that in the UK, some prices adjust faster than New Keynesian model assumes in particular, energy and food which resulted to inflation responding quicker than

expected. This caused interest rates to rise but, its effect on GDP was weak and limited. As the UK is an inflation-targeting country where expectations are well anchored and monetary policy reacts swifty to inflationary pressures meaning that fiscal multipliers on output are likely to be small and short-lived. Though the inverse relationship between GDP and inflation seems counterintuitive at first glance this may reflect productivity led growth.

During crisis periods, a reidentified SVAR structure showed that GDP shocks become more exogenous with fiscal and monetary authorities taking a reactive stance. Government expenditure was shown to respond to output shocks, and inflationary effect from fiscal expansion were more pronounced. Interest rates responded positively to both expenditure and inflationary shocks, reaffirming the role of monetary policy in stabilising prices during economic downturns. The continued negative contemporaneous relationships between inflation and interest ratees during crises suggests a deliberate tolerance of short-term inflation in favour of supporting recovery. Further research could build upon this work by using a time-varying parameter to assess how variables evolve under different macroeconomic conditions.

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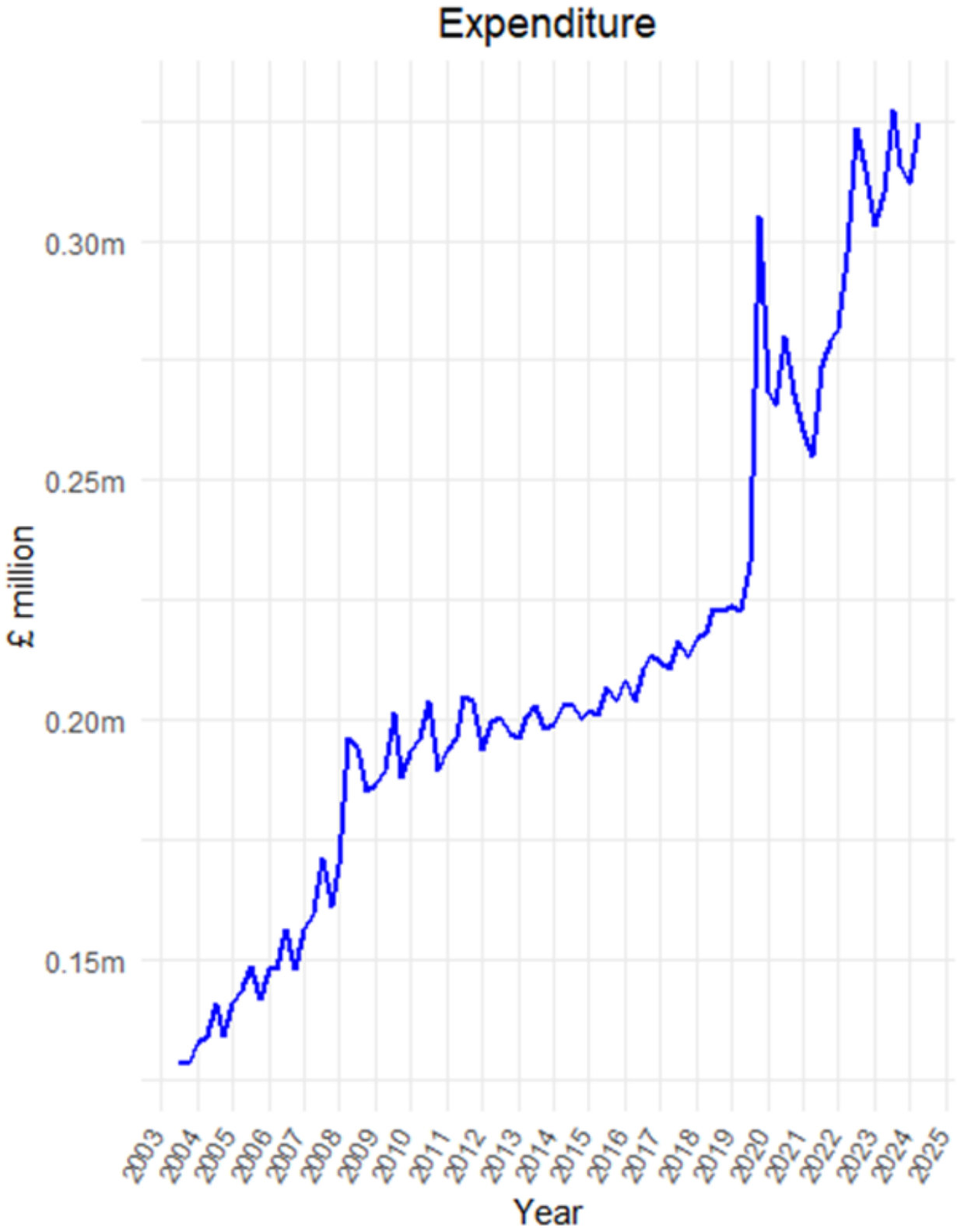
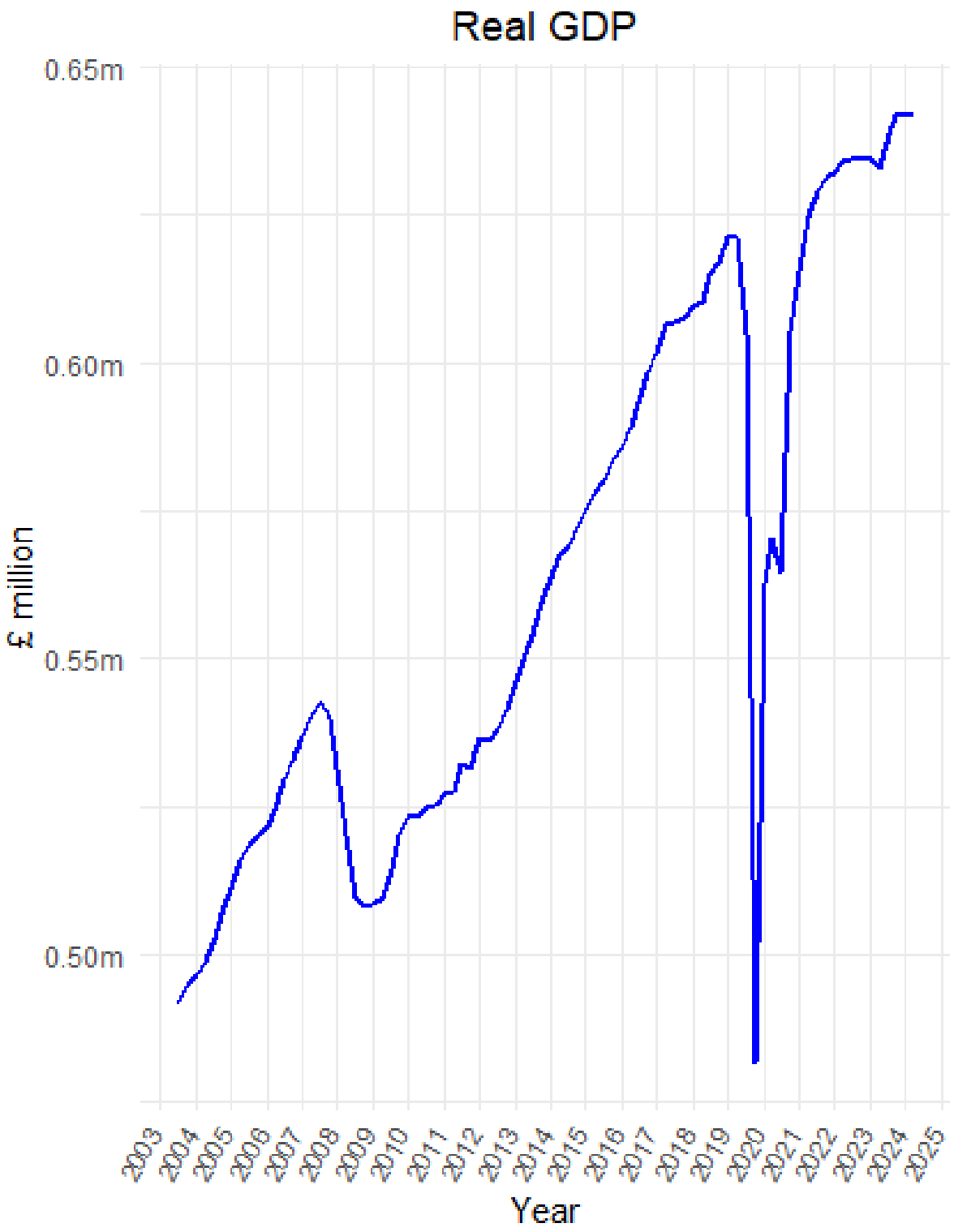
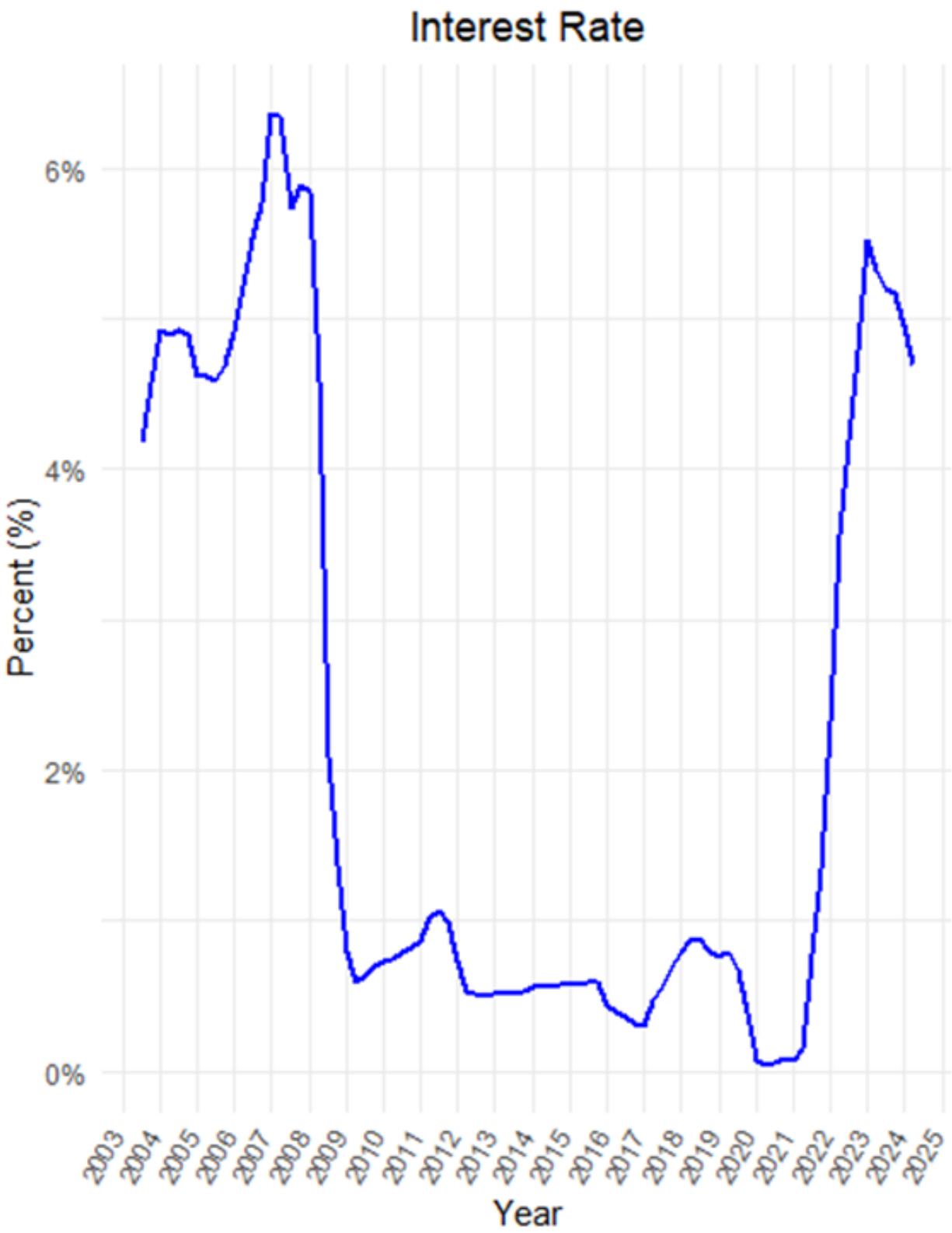
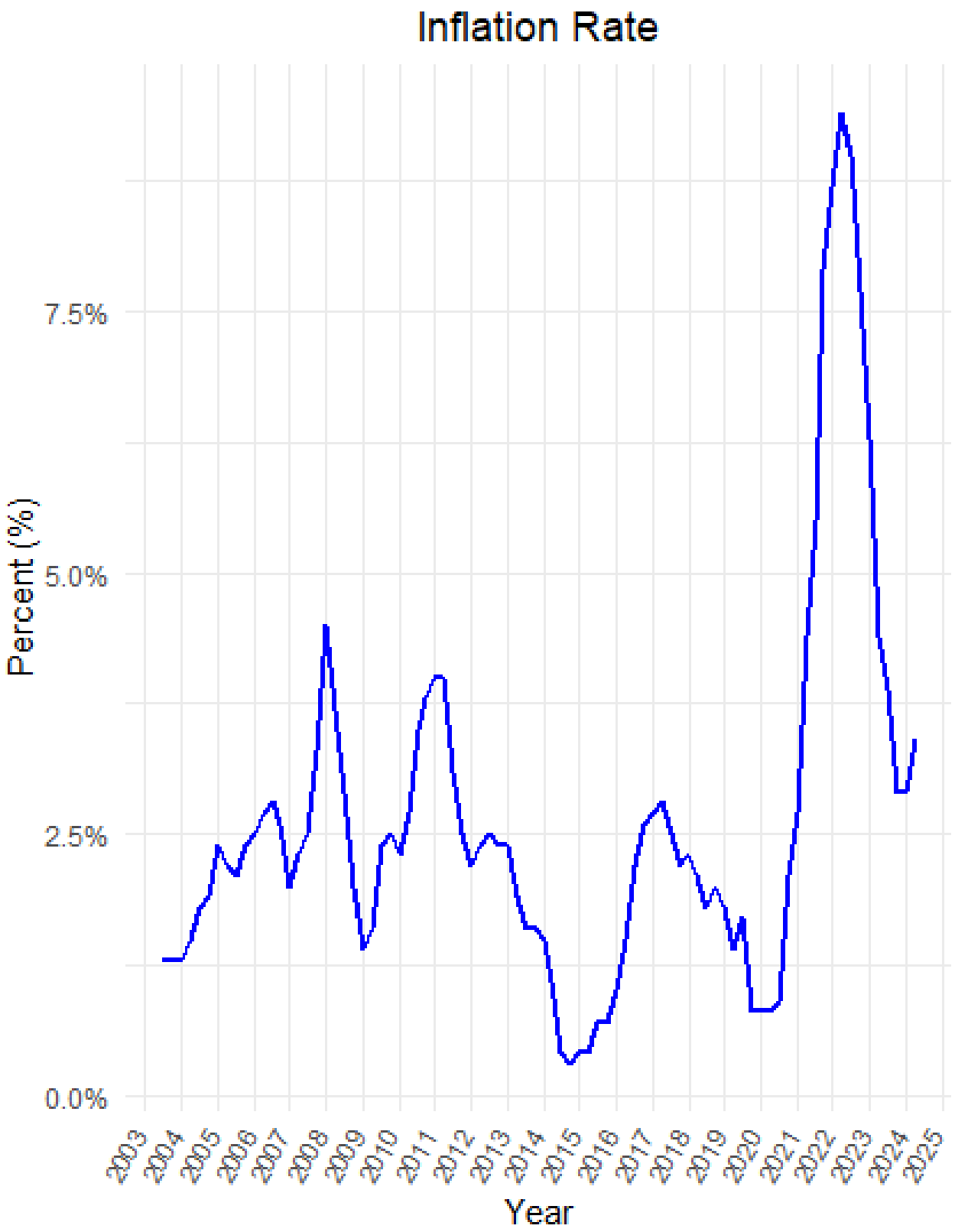
9. Appendices

Appendix 1: Raw Variables




Real GDP
0.65m
0.60m
0.55m
0.50m
£ million
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

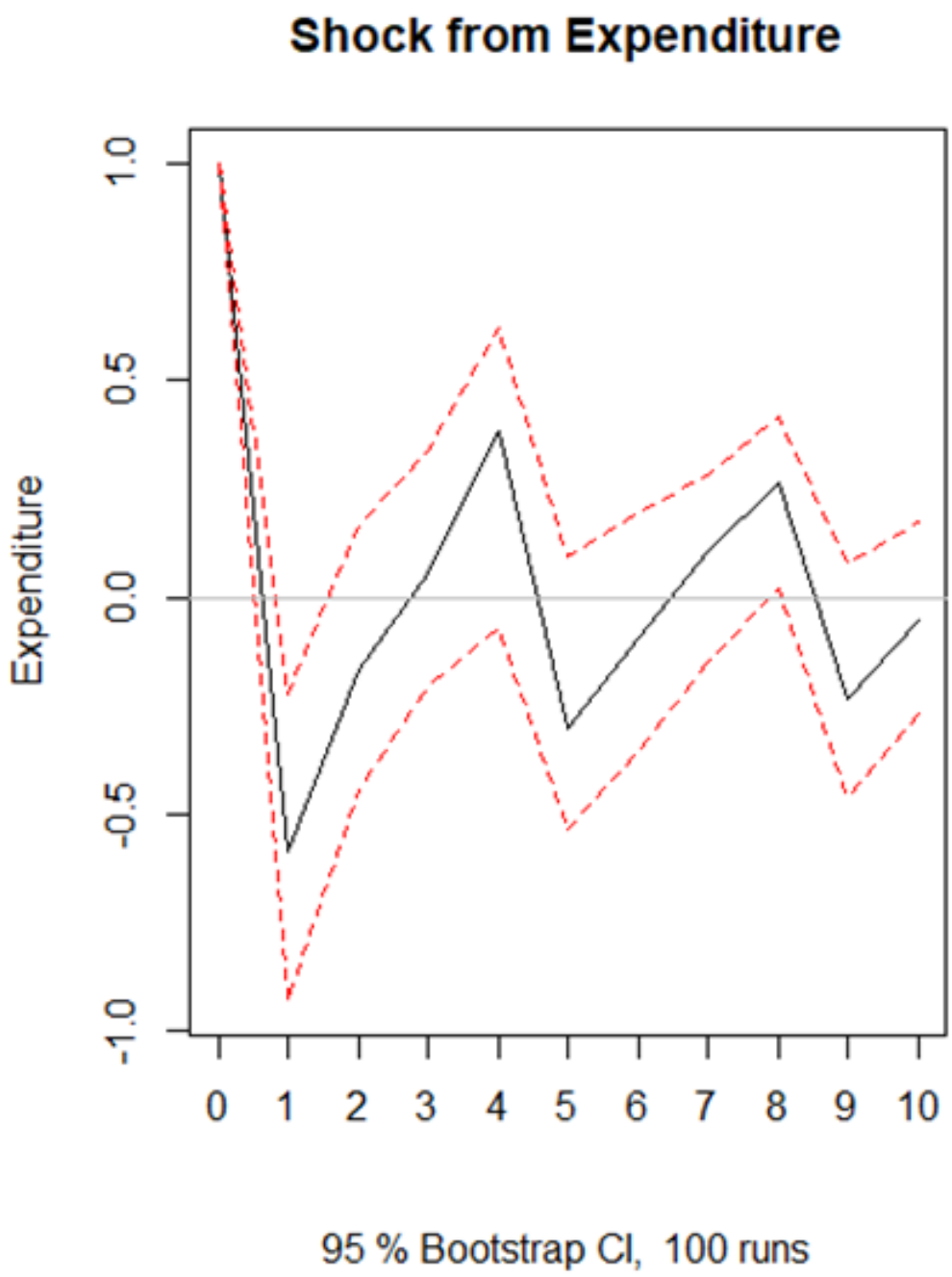

Interest Rate
6%
4%
2%
0%
Percent (%)
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

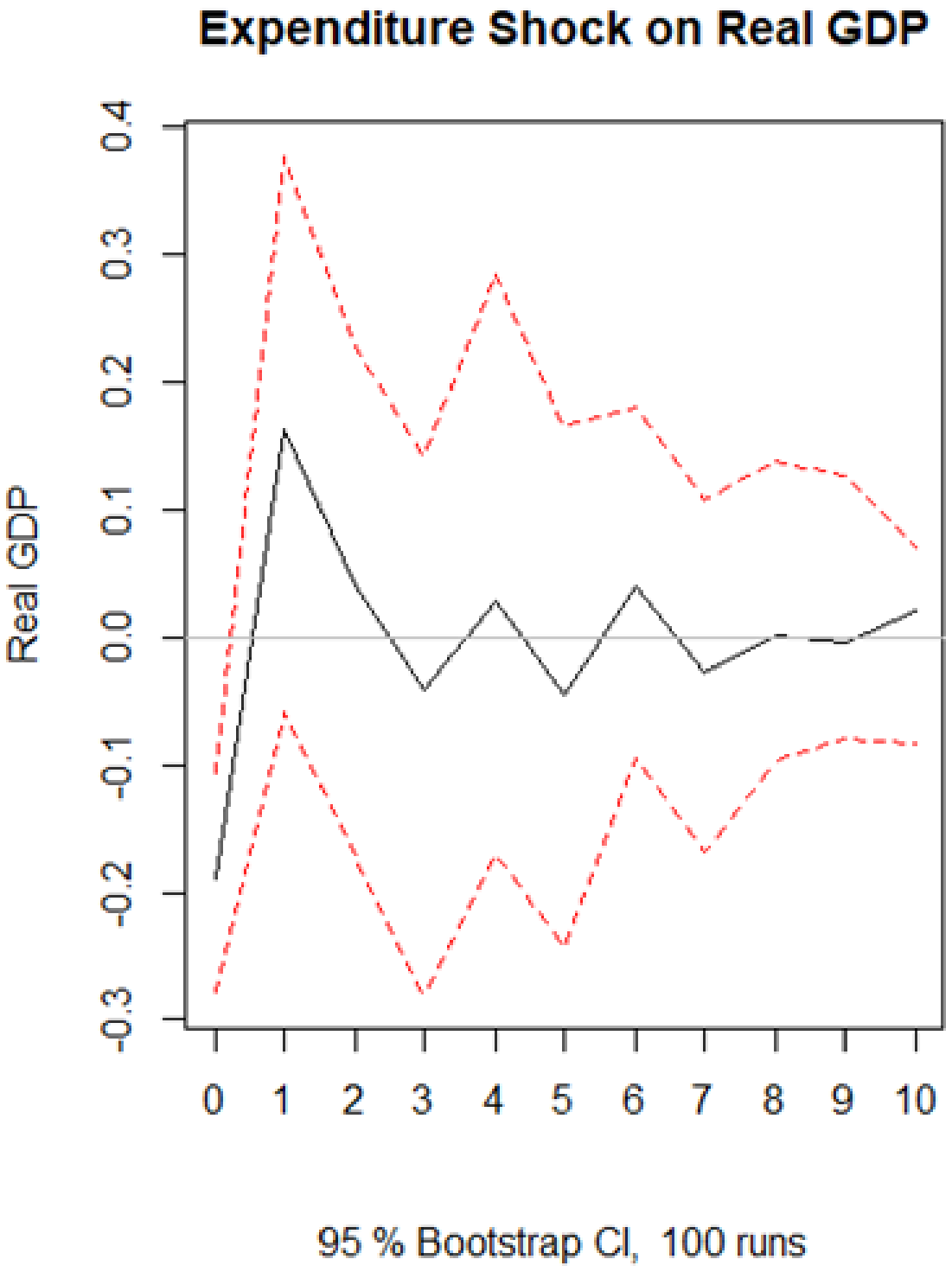

Inflation Rate
Percent (%)
7.5%
5.0%
2.5%
0.0%
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

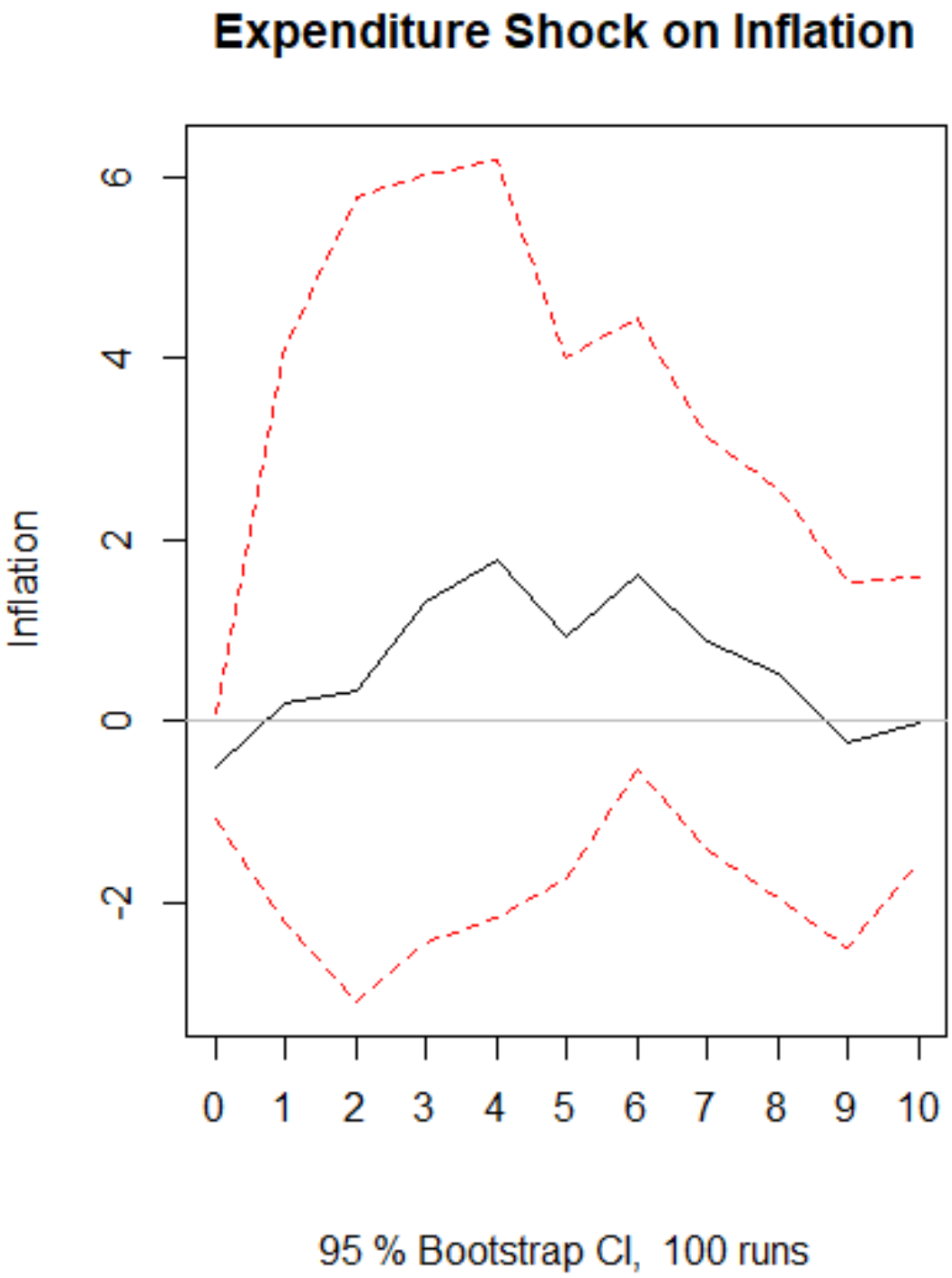
Appendix 2: Summary Statistics

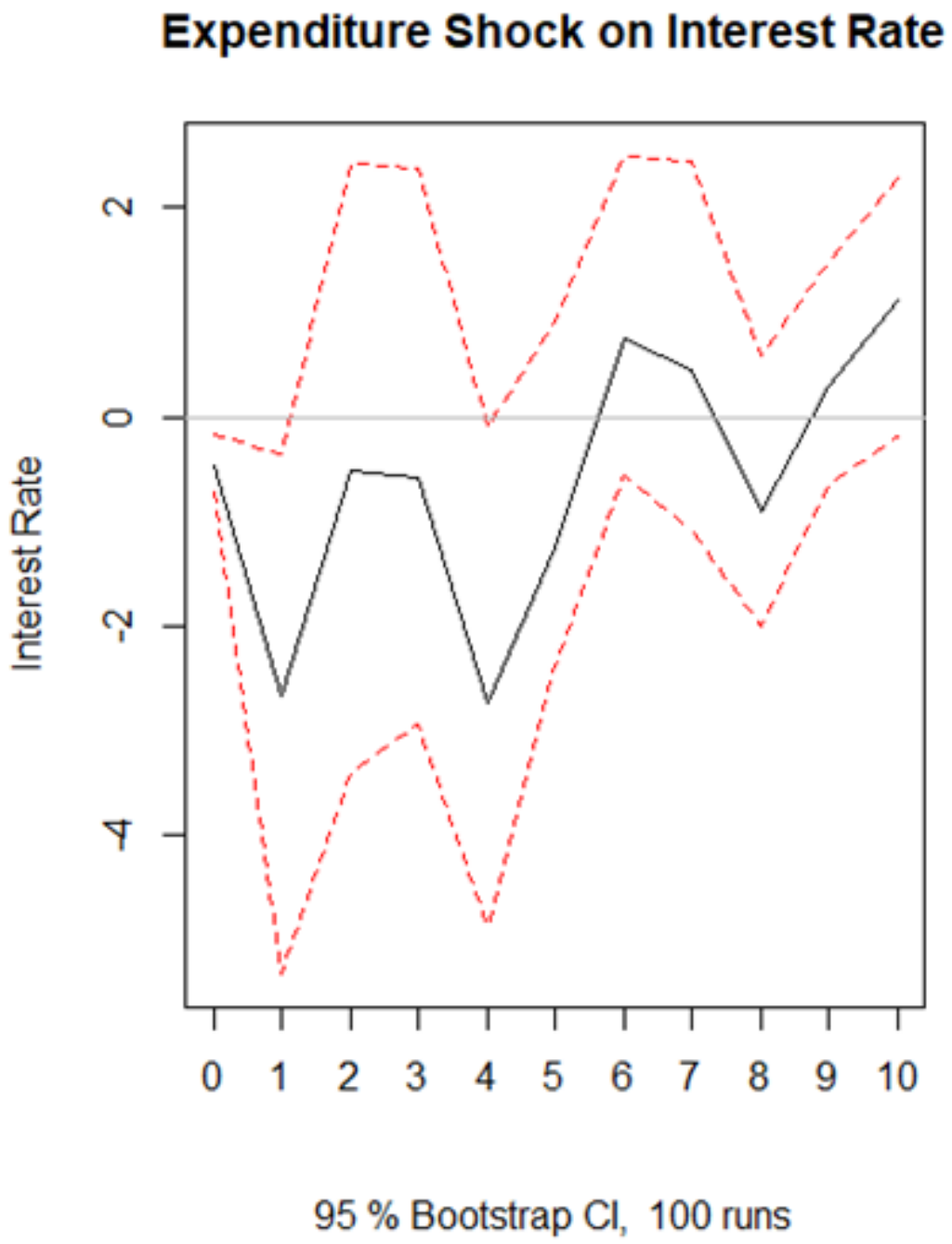
Variable	Mean	Median	SD	Min	Max	Obs
Government Spending	211094.26	202326.50	52240.07	128212.00	327371.00	84
Real GDP	563789.21	556757.00	46846.35	481769.00	642287.00	84
Inflation	2.66	2.35	1.86	0.30	9.40	84
Interest Rate	2.18	0.79	2.16	0.04	6.36	84

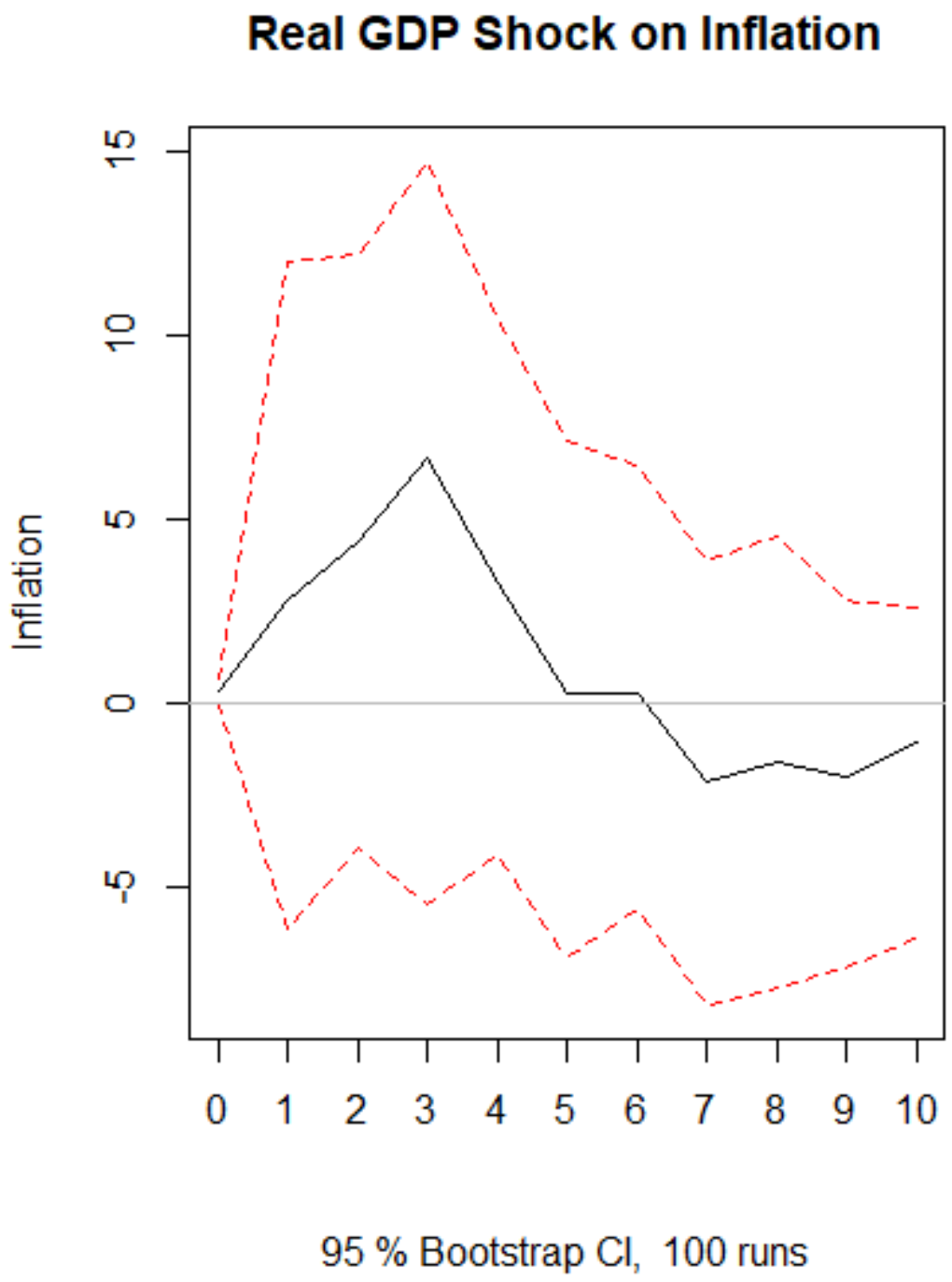
Appendix 3: Structural Impulse Response Functions

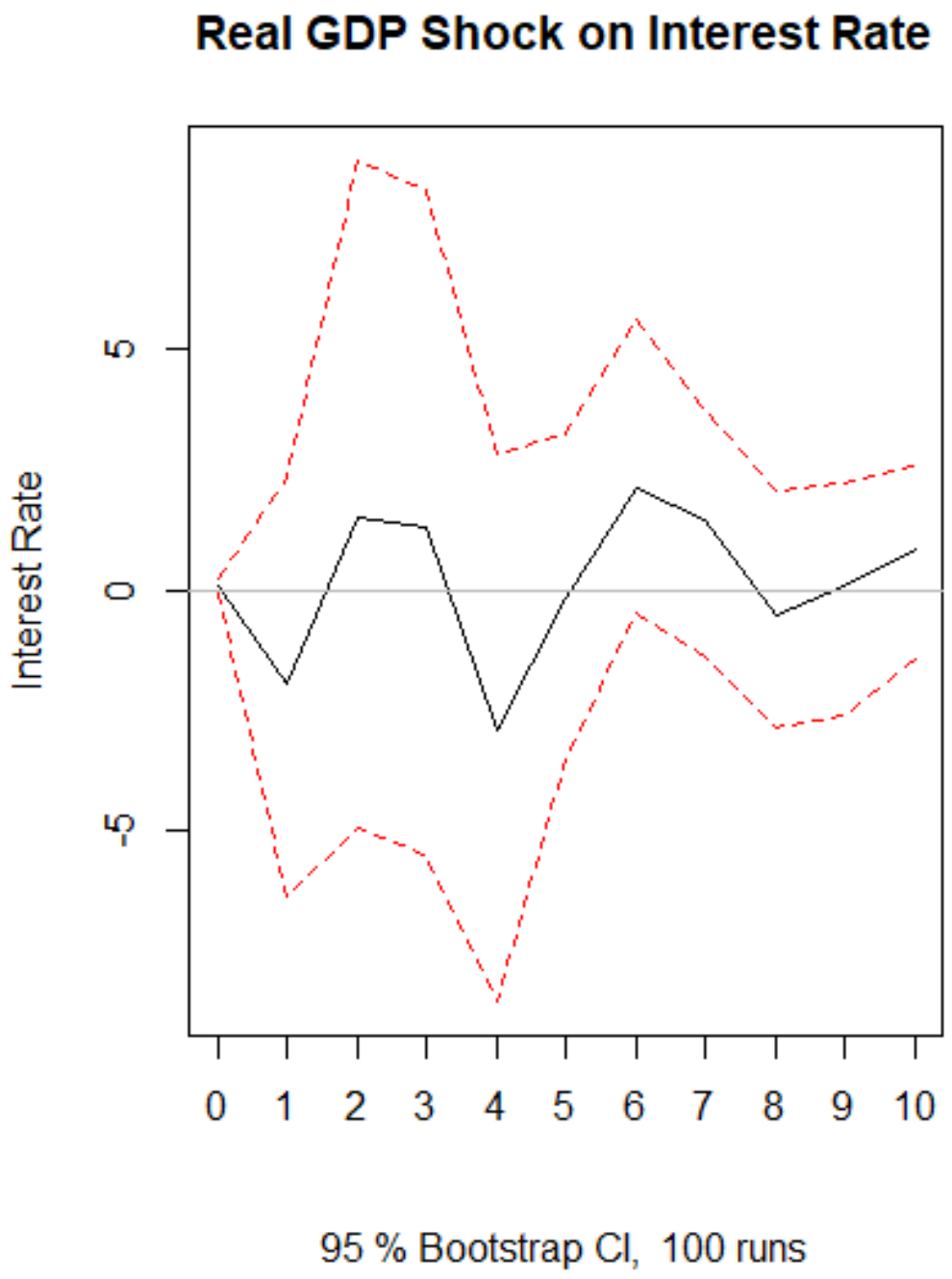


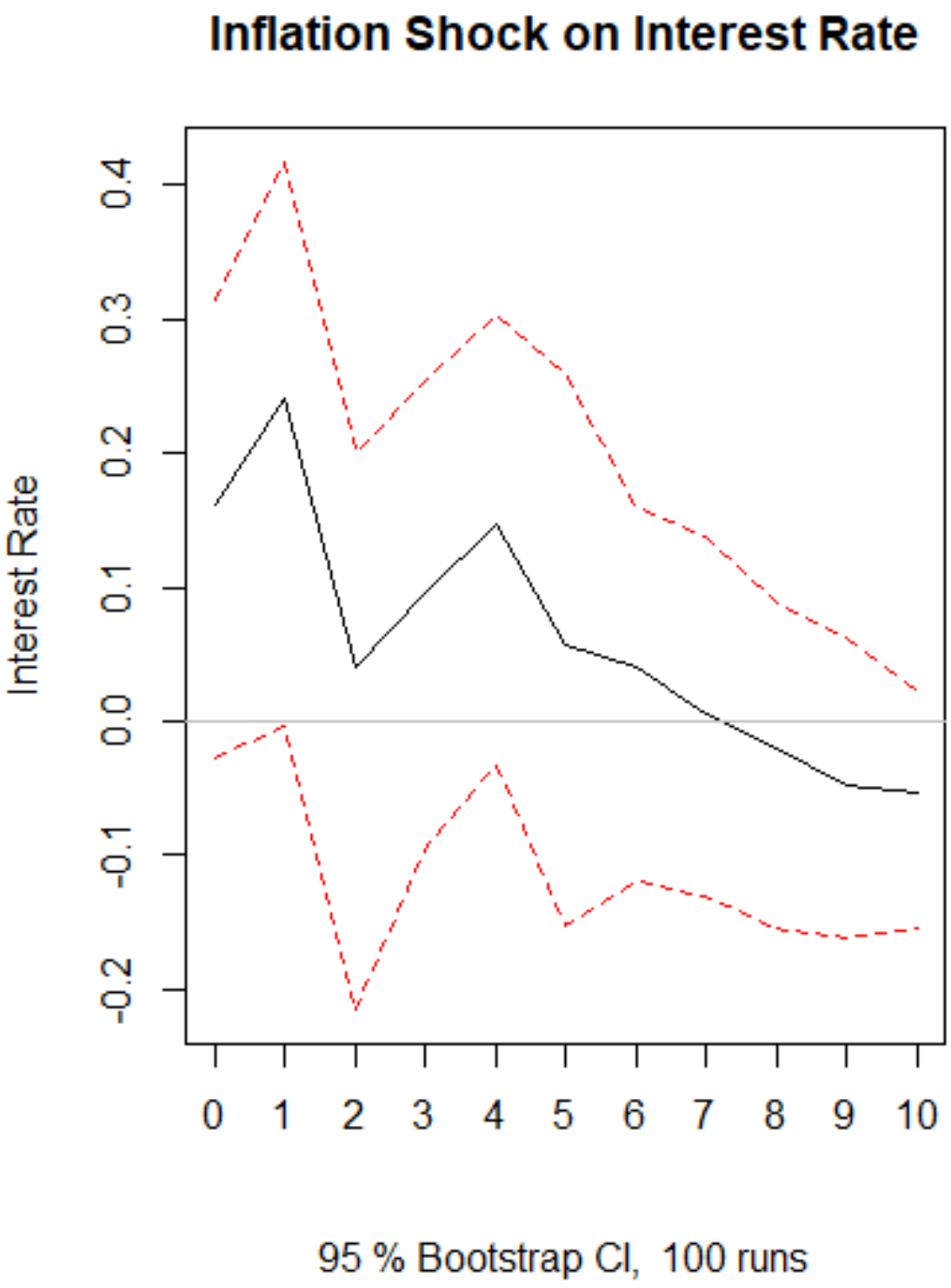

Expenditure Shock on Real GDP
Real GDP
0.4
0.3
0.2
0.1
0.0
-0.1
-0.2
-0.3
0
1
2
3
4
5
6
7
8
9
10
95 % Bootstrap CI, 100 runs


Expenditure Shock on Inflation
Inflation
6
4
2
0
-2
0
1
2
3
4
5
6
7
8
9
10
95 % Bootstrap CI, 100 runs


Expenditure Shock on Interest Rate
Interest Rate
2
0
-2
-4
0
1
2
3
4
5
6
7
8
9
10
95 % Bootstrap CI, 100 runs


Real GDP Shock on Inflation
Inflation
15
10
5
0
-5
0
1
2
3
4
5
6
7
8
9
10
95 % Bootstrap CI, 100 runs


Real GDP Shock on Interest Rate
Interest Rate
5
0
-5
0
1
2
3
4
5
6
7
8
9
10
95 % Bootstrap CI, 100 runs


Inflation Shock on Interest Rate
Interest Rate
0.4
0.3
0.2
0.1
0.0
-0.1
-0.2
0
1
2
3
4
5
6
7
8
9
10
95 % Bootstrap CI, 100 runs